

ipa	viseme	Example 1	Example 2	Example 3
ħ	12	حُرّ	رَحِمَ	الْبَلَح
s	15	سِرّ	قَسَمَ	الْجَالِس
θ	19	ثَرَى	أَحْدَثَ	الْحَارِث
z	15	زِر	قَزَمَ	الْخُبْز
ðˤ	17	ظَنَّ	أَظْهَرَ	قَبِيط
ð	17	ذَلِكَ	أَذَى	أُنْقِذْ
ɣ	20	غَنَى	أَذْمَعَهُ	دِمَاغ
x	12	خَبَرٌ	رَخُوْ	أَخْ
ʃ	16	شَاقَتْ	خَشِيَهُ	وَحْش
sˤ	15	صِلَةٌ	وَصَلَ	غَوْص
j	6	يَوْمٌ	أَذْوَبَةٌ	يَمْدِيرِيْ
w	7	وَلَدٌ	لَوْنٌ	مَكاو
l	14	لَيْسَ	عَلِمَ	عَمَل
m	21	مَجْدٌ	تَمَنُ	قَلَم
n	19	نَائِبٌ	أَذْيَانًا	أَمْن
r	13	رَكْبٌ	قَرْنٌ	يَمْكَرَر
ʕ	12	عَامًا	لَعِبُ	بَيْع

bg-BG

Vowels for bg-BG

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
i	6	искат	сценаристката	лечими
ɛ	4	елен	цена	оспорените

ipa	viseme	Example 1	Example 2	Example 3
ɔ	3	оправя	което	колело
a	2	ангел	докосват	цена
u	7	уклончивата	хубавичка	табу
ja	6,2	ябълка	оял	залая
ɣ	1	ъгъл	ъгъл	имамбаялдъ
ju	6,7	ютя	отвоювал	стою

Consonant for bg-BG

Expand table

ipa	viseme	Example 1	Example 2	Example 3
n	19	неразбираемия	преразпределените	искан
ʒ	16	жълт	тъжен	таралеж
k	20	камък	вкарвал	петък
ʦs	19,15	цар	меценат	царевец
t	19	тайна	натъжени	прост
p	21	приказката	натъпкан	прилеп
r	13	разлика	прозорец	хитър
s	15	сърна	месец	процес
d	19	дом	меден	джаред
x	12	хора	доход	цех
zʲ	15	зян	Замразявайки	
lʲ	14	любов	влюбване	
l	14	лебед	блед	хмел
nʲ	19	някога	понякога	
v	18	време	навреме	лъвов

ipa	viseme	Example 1	Example 2	Example 3
m	21	море	време	корем
b	21	баба	риба	ястреб
g	20	гарван	наруган	драг
dʒ	19,16	Джорджева	тютюнджийския	пледж
f	18	филм	реформа	релеф
m ^j	21	мях	смях	
t ^j	19	тяхна	стяга	
r ^j	13	рядка	впрягайки	
p ^j	21	пяхме	спя	
d ^j	19	дядовите	задявайки	
j	6	йод	пейо	ручей
v ^j	18	вяло	посивяло	
s ^j	15	сякаш	всяка	
b ^j	21	бяла	Избягваме	
k ^j	20	кюпа	прокюрдският	
g ^j	20	гюргево	панагюрското	
f ^j	18	фючърс	Парфюмерия	
z	15	зима	изразходват	разказ
ʃ	16	шума	машина	изпечеш
tʃ	19,16	чак	случаи	меч
dʒ	19,15	дзифт	скръндза	Лодз

ca-ES

Vowels for ca-ES

ipa	viseme	Example 1	Example 2	Example 3
a	2	amen	amaro	està
ɔ	3	odre	ofertori	microtò
ə	1	estan	seré	aigua
e	4	érem	feta	seré
ɛ	4	ecosistema	incorrecta	haver
i	6	itinerants	itinerants	zombi
o	8	ombra	retondre	omissió
u	7	universitaris	candidatures	crono

Consonant for ca-ES

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
b	21	baba	dobla	
β	21	vià	baba	
tʃ	19,16	txadià	matxucs	faig
d	19	dedicada	conduïa	navidad
ð	17	The_Sun	dedicada	trinidad
f	18	facilitades	afectarà	àgraf
g	20	gracia	congratula	
ɣ	20		aigua	
j	6	hiena	esplaia	cofoi
dʒ	19,16	djakarta	compostatge	george
k	20	curós	dodecà	doblec
l	14	laberint	miolar	preval
ʎ	14	lligada	millorarà	perbull

ipa	viseme	Example 1	Example 2	Example 3
m	21	macadàmies	femar	sublim
n	19	necessaris	sanitaris	alterament
ŋ	20		algonquí	albenc
ɲ	19	nyasa	remenjar	alemany
p	21	pegues	estepa	cap
ɾ	19		caro	càrter
r	13	rabada	carro	lofòfor
s	15	ceri	cursar	cus
ʃ	16	xacar	microxip	midraix
t	19	tabacaires	estratifica	debatut
θ	19	ceará	vecinos	Álvarez
w	7	westfalià	inaugurar	inscriu
x	12	juanita	mujeres	heinrich
z	15	zelar	brasils	alianze
ʒ	16	gebrada	ascogènic	orange

cs-CZ

Vowels for cs-CZ

Expand table

ipa	viseme	Example 1	Example 2	Example 3
ɪ	6	ideální	handicapované	giganty
ɛ	4	efektní	grotesce	hygieně
a	2	agens	infarktu	extra
o	8	oáz	hřbitovy	čisto

ipa	viseme	Example 1	Example 2	Example 3
u	7	ubrání	komponuje	grilu
i:	6	íránské	jedinými	generační
ɛ:	4	éry	jednoduchého	gumové
a:	2	árijské	brán	hořká
o:	8	ódu	famózního	jó
u:	7	úbočí	petúnie	grilů
oʊ	8,4	ouha	fanouška	hanbou
aʊ	2,4	auto	hydrauliky	
ɛʊ	4,4	euforie	terapeutická	
ə	1			

Consonant for cs-CZ

[Expand table](#)

ipa	viseme	Example 1	Example 2	Example 3
p	21	padá	hyperaktivní	handicap
b	21	balon	hraběte	
t	19	tabák	étos	agent
d	19	disco	čidel	
c	16	tuhýk	kandidáti	kaprad'
ʃ	16	d'ábel	hrdiny	
k	20	kaligrafie	důkazní	grafik
g	20	gangstera	hygienici	
ʦ	19,15	civilistům	evoluce	klávesnic
ʤ	19,15		leckde	
tʃ	19,16	čepice	hlídači	klíč

ipa	viseme	Example 1	Example 2	Example 3
ɔ̃	19,16	džusy	kilojouľ	
f	18	fádní	grafice	graf
v	18	vabank	exkluzivita	
s	15	sekundě	grimasami	impuls
z	15	záhadným	gruzínské	
ɾ	13	řad	hořícím	
ʃ	16	šálků	grošů	kěž
ʒ	16	žab	loupeže	
j	6	já	čihající	línej
x	12	chlapcem	lichotí	domorodých
ɦ	12	hektar	historického	
r	13	rabat	guru	faktor
l	14	ladu	eliminovat	netajil
m	21	macešsky	amatérský	cením
n	19	následkům	cenné	gurmán
ŋ	20		kolonky	
ɲ	19	ňadra	komorně	kolegyň
ɱ	21		komfort	
ɹ̥	13	třída	extratřídý	komentář

da-DK

Vowels for da-DK

[Expand table](#)

ipa	viseme	Example 1	Example 2	Example 3
a	2	abitur	kortfattet	marina
ɑ	2	abnorme	picaresk	varer
ɑ:	2	ara	pirat	vandkar
ɛ	4	edderkop	andægtig	tomatpuré
ɛ:	4	eventyr	dyrlæge	
ɔ	3	onde	dysfunktion	Suså
ɒ	2	ånd	abehand	akkumulator
ɒ:	2	årbog	fyrreårig	rickshaw
ɔ:	3	åbne	modstående	husarblå
e	4		buntmagere	bortgifter
æ:	1	abegilde	flagdage	hovedfag
e	4	editere	klarheden	kanapé
ø	1	ødem	mælkebøtte	miljø
ø:	1	øde	bortførere	lindø
ə	1	Eyolf	meneders	æde
e:	4	edelweiss	placebo	Culmsee
i	6	iaften	lipizzaner	bankkonti
i:	6	ilinger	industrien	bagi
o	8	oase	udelød	bravo
œ	4	øm	varmerør	bøh
œ:	4	Earl	arbejdsprøves	Edinburgh
o:	8	oberst	bevatrons	congo
u	7	udøver	spektakulær	endnu
u:	7	uge	spidsbue	eisuu
y	4	ybsalon	underkrydder	menu

ipa	viseme	Example 1	Example 2	Example 3
y:	4	yde	vidsyn	dødssyg

Consonant for da-DK

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
b	21	blandt	gabende	dåb
d	19	då	hævder	grædt
ð	17	thousand	baaden	grad
f	18	fabrik	spøgefugle	boligstof
g	20	gaader	fåborgenser	fik
h	12	håb	trægheden	
j	6	hjorte	miljø	bagtøj
k ^h	20	kabale	bortkomme	Cuc
l	14	lå	romanblad	fatal
m	21	maya	taxametret	calcium
n	19	nå	togene	bredden
ŋ	20		funktion	klarering
p ^h	21	pa	culpa	
ʔ	19		affyre	bortgå
ʁ	13	rå	areal	
ɟ	4		jonglørkunst	bjørnebær
s	15	så	person	belønnes
ʃ	16	Sjælland	benzinstation	affiche
t	19	toge	sidetal	administrationsapparat
v	18	vaagner	evaluere	dåkalv

ipa	viseme	Example 1	Example 2	Example 3
w	7	walkman	prøve	meterlov

de-DE/de-CH/de-AT

Suprasegmentals for German

 Expand table

Example 1 (Onset for consonant, word-initial for vowel)	Example 2 (Intervocalic for consonant, word-medial nucleus for vowel)	Example 3 (Coda for consonant, word-final for vowel)	Comments
anders /a 1 n - d ax r s/	Multiplikationszeichen /m uh l - t iy - p l iy - k a - ts y ow 1 n s - ts ay - c n/	Biologie /b iy - ow - l ow - g iy 1/	Speech service phone set put stress after the vowel of the stressed syllable
Allgemeinwissen /a 2 l - g ax - m ay 1 n - v ih - s n/	Abfallentsorgungsfirma /a 1 p - f a l - ^ eh n t - z oh 2 ax r - g uh ng s - f ih ax r - m a/	Computertomographie /k oh m - p y uw 1 - t ax r - t ow - m ow - g r a - f iy 2/	The Speech service phone set puts stress after the vowel of the sub-stressed syllable

Vowels for German

 Expand table

sapi	ipa	VisemeID	Example 1	Example 2	Example 3
a:	a:	2	Aber	Maßstab	Schema
a	a	2	Abfall	Bach	Agatha
oh	ɔ	3	Osten	Pfosten	
eh:	ɛ:	4	Ähnlichkeit	Bär	Fasciae ¹
eh	ɛ	4	ändern	Prozent	Amygdalae
ax	ə	1	'verstauen ²	Aachen	Frage

sapi	ipa	VisemeID	Example 1	Example 2	Example 3
iy	i:	6	Iran	abbiegt	Relativitätstheorie
ih	ɪ	6	Innung	singen	Woody
eu	ø:	1	Ösen	ablösen	Malmö
ow	o, o:	8	ohne	Balkon	Treptow
oe	œ	4	Öffnung	befördern	
ey	e, e:	4	Eberhard	abfegt	b
uw	u:	7	Udo	Hut	Akku
uh	ʊ	4	Unterschiedes	bunt	
ue	y:	4	Übermut	pflügt	Menü
uy	ʏ	7	üppig	System	

- 1 Only in words of foreign origin, such as *Fasciae*.
- 2 Word-initial only in words of foreign origin, such as *Appointment*. Syllable-initial in 'verstauen.

Diphthong for German

[Expand table](#)

sapi	ipa	VisemeID	Example 1	Example 2	Example 3
ay	aɪ	2,6	einsam	Unabhängigkeit	Abtei
aw	au	2,7	außen	abbaust	Stau
oy	ɔy, ɔʏ	3,4	Euphorie	träumt	scheu

Semivowels for German

[Expand table](#)

sapi	ipa	VisemeID	Example 1	Example 2	Example 3
ax r	ɐ	4		abändern	locker

Consonants for German

sapi	ipa	VisemeID	Example 1	Example 2	Example 3
b	b	21	Bank		Pub ¹
d	d	19	danken	Lendl ²	Claude ³
jh	ɖ	16	Jeff	gemanagt	Change ⁴
f	f	18	Fahrtdauer	angriffslustig	abbruchreif
g	g	20	gut	Greg ⁵	
h	h	12	Hausanbau		
y	j	6	Jod	Reaktion	hui
k	k	20	Koma	Aspekt	Fleck
l	l	14	lau	ähneln	zuviel
m	m	21	Mut	Amt	Lehm
n	n	19	nun	und	Huhn
ng	ŋ	20	Nguyen ⁶	Schwank	Ring
p	p	21	Partner	abrupt	Tip
pf	pf	21,18	Pferd	dampft	Topf
r	ʀ, r, ʁ	13	Reise	knurrt	Haar
s	s	15	Staccato ⁷	bist	mies
sh	ʃ	16	Schule	mischt	lappisch
t	t	19	Traum	Straße	Mut
ts	ts	19,15	Zug	Arzt	Witz
ch	tʃ	19,16	Tschechien	aufgeputscht	bundesdeutsch
v	v	18	winken	Qualle	Groove ⁸
x	x ⁹ , ç ¹⁰	12	Bacherach ¹¹	Macht möglichst	Schmach 'ich
z	z	15	super		
zh	ʒ	16	Genre	Breezinski	Edvige

- 1 Only in words of foreign origin, such as *Pub*.
- 2 Only in words of foreign origin, such as *Lendl*.
- 3 Only in words of foreign origin, such as *Claude*.
- 4 Only in words of foreign origin, such as *Change*.
- 5 Word-terminally only in words of foreign origin, such as *Greg*.
- 6 Only in words of foreign origin, such as *Nguyen*.
- 7 Only in words of foreign origin, such as *Staccato*.
- 8 Only in words of foreign origin, such as *Groove*.
- 9 The IPA **x** is a hard "ch" after all non-front vowels (a, aa, oh, ow, uh, uw, and the diphthong aw).
- 10 The IPA **ç** is a soft "ch" after front vowels (ih, iy, eh, ae, uy, ue, oe, eu, and diphthongs ay, oy) and consonants.
- 11 Word-initial only in words of foreign origin, such as *Juan*. Syllable-initial also in words such as *Bacherach*.

Oral consonants for German

[Expand table](#)

sapi	ipa	VisemeID	Example
^	ʔ	19	beachtlich /b ax - ^ a 1 x t - l ih c/

⚠ Note

We need to add a [gs] phone between two distinct vowels, except when the two vowels are a genuine diphthong. This oral consonant is a glottal stop. For more information, see [glottal stop](#). Besides, de-CH, de-AT locales don't support SAPI phones now.

el-GR

Vowels for el-GR

[Expand table](#)

ipa	viseme	Example 1	Example 2	Example 3
a	2	άκουσμα	μαγαζάκι	ουδέτερα
e	4	αίτιο	κατέχω	ουδέποτε

ipa	viseme	Example 1	Example 2	Example 3
i	6	εισπνοή	ξυρισμένο	εαρινή
o	8	όαση	δώρο	δυ**ο
u	7	συγγρικό	παπούτσι	μαϊμού

Consonant for el-GR

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
b	21	μπορώ	φάμπρικα	παμπ
c	16	και	σακί	
ç	12	χέρι	μοναχή	
d	19	ντύνει	πέντε	οφ-σάιντ
ð	17	δρόμος	κραδασμοί	
ɖz	19,15	τζάμι	φλάντζα	
f	18	φεύγω	καφετέρια	ουφ
g	20	γκρέμισε	όγκος	πινγκ-πονγκ
ɣ	20	γαστρονομική	μεγαλόπνοα	
ʒ	16	γκισέ	φαράγγι	
j	6			
ɟ	12	γέρος	κραγιόν	
k	20	κάρτα	ιππικό	κρακ
l	14	λόγος	μιλώ	προφίλ
m	21	μιλώ	δραματική	πριμ
n	19	νόμιμο	Δανέζα	πριν
p	21	πίνω	απέτυχαν	σελοτέιπ
r	19	ρούχα	εαρινή	σέντερ

ipa	viseme	Example 1	Example 2	Example 3
s	15	σελίδα	μέσο	πλάτος
t	19	τότε	τότε	κιτ
θ	19	θέλω	φορολογηθείς	μαμούθ
ts	19,15	τσάγια	χαράτσι	ματς
v	18	βράδυ	διαβάζω	μοβ
x	12	χρόνος	μονάχα	αχ
z	15	ζέστη	χιονίζει	πλαζ

en-GB/en-IE/en-AU

Vowels for en-GB

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
ɑː	2		fast	bra
æ	1		fat	
ʌ	1		bug	
ɛə	4,1			hair
aʊ	2,4	out	mouth	how
ə	1	a		driver
aɪ	2,6		five	
ɛ	4	egg	dress	
ɜː	5	ernest	shirt	fur
eɪ	4,6	ailment	lake	pay
ɪ	6		adding	
ɪə	6,1		beard	hear

ipa	viseme	Example 1	Example 2	Example 3
i:	6	eat	seed	see
ɒ	2		pod	
ɔ:	3		dawn	
əʊ	1,4		code	pillow
ɔɪ	3,6		point	boy
ʊ	4		look	
ʊə	4,1			tour
u:	7		food	two

Consonant for en-GB

[↗](#) Expand table

ipa	viseme	Example 1	Example 2	Example 3
b	21	bike	ribbon	rib
tʃ	19,16	challenge	nature	rich
d	19	date	caddy	slid
ð	17	this	father	breathe
f	18	face	laughing	enough
g	20	gold	bragging	beg
h	12	hurry	ahead	
j	6	yes		
dʒ	19,16	gin	badger	bridge
k	20	cat	lucky	truck
l	14	left	gallon	fill
m	21	mile	limit	ham
n	19	nose	phonetic	tin

ipa	viseme	Example 1	Example 2	Example 3
ŋ	20		singer	long
p	21	price	super	tip
ɹ	13	rate	very	
s	15	say	sissy	pass
ʃ	16	shop	cashier	leash
t	19	top	kitten	bet
θ	19	theatre	mathematics	breath
v	18	very	liver	have
w	7	will		
z	15	zero	blizzard	rose
ʒ	16		vision	beige

en-US/en-CA

Suprasegmentals for English

 Expand table

Example 1 (onset for consonant, word-initial for vowel)	Example 2 (intervocalic for consonant, word-medial nucleus for vowel)	Example 3 (coda for consonant, word-final for vowel)	Comments
burger /b er 1 r - g ax r/	falafel /f ax - l aa 1 - f ax l/	guitar /g ih - t aa 1 r/	The Speech service phone set puts stress after the vowel of the stressed syllable.
inopportune /ih 2 - n aa - p ax r - t uw 1 n/	dissimilarity /d ih - s ih 2- m ax - l eh 1 - r ax - t iy/	workforce /w er 1 r k - f ao 2 r s/	The Speech service phone set puts stress after the vowel of the sub-stressed syllable.

Vowels for English

[Expand table](#)

sapi	ipa	VisemeID	Example 1	Example 2	Example 3
iy	i	6	eat	feel	valley
ih	ɪ	6	if	fill	
ey	eɪ	4,6	ate	gate	day
eh	ɛ	4	every	pet	meh (rare word-final)
ae	æ	1	active	cat	nah (rare word-final)
aa	ɑ	2	obstinate	poppy	rah (rare word-final)
ao	ɔ	3	orange	cause	Utah
uh	ʊ	4	book		
ow	oʊ	8,4	old	clone	go
uw	u	7	Uber	boost	too
ah	ʌ	1	uncle	cut	
ay	aɪ	11	ice	bite	fly
aw	aʊ	9	out	south	cow
oy	ɔɪ	10	oil	join	toy
y uw	ju	6,7	Yuma	human	few
ax	ə	1	ago	woman	area

R-colored vowels for English

[Expand table](#)

sapi	ipa	VisemeID	Example 1	Example 2	Example 3
ih r	ɪɹ	6,13	ears	tiramisu	near
eh r	ɛɹ	4,13	airplane	apparently	scare
uh r	ʊɹ	4,13			cure
ay r	aɪɹ	11,13	Ireland	fireplace	choir

sapi	ipa	VisemeID	Example 1	Example 2	Example 3
aw r	auɹ	9,13	hours	powerful	sour
ao r	ɔɹ	3,13	orange	moral	soar
aa r	ɑɹ	2,13	artist	start	car
er r	ə	5	earth	bird	fur
ax r	ə	1		allergy	supper

Semivowels for English

 Expand table

sapi	ipa	VisemeID	Example 1	Example 2	Example 3
w	w	7	with, suede	always	
y	j	6	yard, few	onion	

Aspirated oral stops for English

 Expand table

sapi	ipa	VisemeID	Example 1	Example 2	Example 3
p	p	21	put	happen	flap
b	b	21	big	number	crab
t	t	19	talk	capital	sought
d	d	19	dig	random	rod
k	k	20	cut	slacker	Iraq
g	g	20	go	ago	drag

Nasal stops for English

 Expand table

sapi	ipa	VisemeID	Example 1	Example 2	Example 3
m	m	21	mat, smash	camera	room
n	n	19	no, snow	tent	chicken
ng	ŋ	20		link	sing

Fricatives for English

 Expand table

sapi	ipa	VisemeID	Example 1	Example 2	Example 3
f	f	18	fork	left	half
v	v	18	value	event	love
th	θ	19	thin	empathy	month
dh	ð	17	then	mother	smooth
s	s	15	sit	risk	facts
z	z	15	zap	busy	kids
sh	ʃ	16	she	abbreviation	rush
zh	ʒ	16	Jacques	pleasure	garage
h	h	12	help	enhance	a-ha!

Affricates for English

 Expand table

sapi	ipa	VisemeID	Example 1	Example 2	Example 3
ch	tʃ	19,16	chin	future	attach
jh	dʒ	19,16	joy	original	orange

Approximants for English

 Expand table

sapi	ipa	VisemeID	Example 1	Example 2	Example 3
l	l	14	lid, glad	palace	chill
r	ɹ	13	red, bring	borrow	tar

ⓘ Note

en-CA locale doesn't support SAPI phones.

es-ES

Vowels for es-ES

 Expand table

sapi	ipa	viseme	Example 1	Example 2	Example 3
a	a	2	alto	cantar	casa
i	i	6	ibérica	avispa	taxi
e	e	4	elefante	atento	elefante
o	o	8	ocaso	encontrar	ocaso
u	u	7	usted	punta	Juanlu

Consonant for es-ES

 Expand table

sapi	ipa	viseme	Example 1	Example 2	Example 3
b	b	21	baobab	cambio	amb
	β	21		baobab	baobab
ch	tʃ	19,16	cheque	coche	Marraquech
d	d	19	dedo	candado	portland
	ð	17		dedo	verdad

sapi	ipa	viseme	Example 1	Example 2	Example 3
f	f	18	fácil	elefante	puf
g	g	20	ganga	ganga	dóping
	ɣ	20		agua	tuareg
j	j	6	iodo	caliente	rey
jj	ɟɟ	6,6		villa	
k	k	20	coche	boca	titánic
l	l	14	lápiz	ala	cordel
ll	ʎ	14	llave	conllevar	
m	m	21	morder	amar	álbum
n	n	19	nada	cena	ratón
nj	ɲ	19	ñaña	arañazo	
p	p	21	poca	topo	stop
r	r	19		cara	abrir
rr	rr	13	radio	corre	purr
s	s	15	saco	vaso	pelos
t	t	19	toldo	atar	disquet
th	θ	19	zebra	azul	lápiz
w	w	7	hueso	agua	guau
x	x	12	jota	ajo	reloj

💡 Tip

The **es-ES** Speech service phone set doesn't support the following Spanish IPA: **β**, **ð**, and **ɣ**. If they're needed, consider using the IPA directly.

es-MX

Vowels for es-MX

[Expand table](#)

ipa	VisemeID	Example 1	Example 2	Example 3
a	2	azúcar	tomate	ropa
e	4	eso	remero	amé
i	6	hilo	liquido	olí
o	8	hogar	olote	caso
u	7	uno	ninguno	tabú

Consonants for es-MX

[Expand table](#)

ipa	VisemeID	Example 1	Example 2	Example 3
b	21	bote		
β	21	órbita	envolvente	
tʃ	19,16	chico	hacha	
d	19	dátil		
ð	17	orden	oda	
f	18	foco	oficina	
g	20	gajo		
ɣ	20	agua	hoguera	
j	6	iodo	caliente	rey
ʝ	6,6		olla	
k	20	casa	ácaro	
l	14	loco	ala	
ʎ	14	llave	enyugo	
m	21	mata	amar	

ipa	VisemeID	Example 1	Example 2	Example 3
n	19	nada	ano	
ɲ	19	ñoño	año	
p	21	papa	papa	
ɾ	19		aro	
r	13	rojo	perro	
s	15	silla	asa	
t	19	tomate		soft
w	7	huevo		
x	12	jarra	hoja	

fi-FI

Vowels for fi-FI

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
ɑ	2	avautuu	vaihtuvan	pouta
ɑ̃	2,6	aika	vaihtuu	lauantai
ɑ̃u	2,7	aura	uloskirjaudu	Passau
ɑ:	2	ay-väen	neutraali	poutaa
æ	1	äveriäs	öljyjätin	pöytä
æ̃	1,6	äiti	iäkkäiden	täi
æ̃y	1,4	äyrin	täytyy	käy
æ:	1	ääriryhmiä	häädetään	päivää
e	4	enköhän	terve	me
ẽ	4,6	ei	vaihteita	hei

ipa	viseme	Example 1	Example 2	Example 3
ø	1	öljyalan	ulkonäön	tiedänpö
ø̞	1,6	öisin	töitä	viittilöi
ø̝	1,4	öylätti	pöytä	
ø:	1	Öölanti	ulkoministeriöön	Bodø
eu	4,7	eurot	kyläseura	leu
ey	4,4	Eysturoy	keskeytyä	
e:	4	eesti	kyljelleen	aiheuttanee
i	6	iäkästä	viha	Berliini
i̞e	6,4	ientaskun	kieli	lie
i̞u	6,7		viulu	
i̞y	6,4		vihkiytynyt	
i:	6	iida	siika	solmii
o	8	oksa	asuintaloja	spekulaatio
o̞i	8,6	oivia	koittaa	spekuloi
o̞u	8,7	outo	autokoulu	window
o:	8	ok	koostaa	yo
u	7	ufoista	Bärlund	jättemaksu
u̞i	7,6	ui	muuta	epäonnistui
u̞o	7,8	Uolevi	Suomi	Hilavuo
u:	7	url	innokkuus	kiikkuu
y	4	ydin	ökyrikas	kesy
yø	4,1	yö	työtä	järjestötyö
y̞i	4,6	Yichangin	syitä	järjestäytyi
y:	4	yo	ryyppy	iskeytyy

Consonant for fi-FI

ipa	viseme	Example 1	Example 2	Example 3
b	21	baareissa	Urban	Jakob
d	19	dementia	ladot	jugend
f	18	face	afgaani	Alf
g	20	gaalassa	fagotti	Aslög
h	12	ha	astumahan	Dietrich
j	6	jää	öljyä	Kaj
k	20	Kajaanin	epätarkat	idyllic
l	14	Lyytikäinen	euroseteleiden	mail
m	21	mä	roimat	späm
n	19	nää	baanalle	iäkkään
ŋ	20		nähdäänkin	planning
p	21	pa	epäsuoria	backup
r	13	risteilyn	baari	Player
s	15	sä	öljyisiä	Bärnäs
ʃ	16	Schauman	Bangladeshin	cash
t	19	tä	eurosta	epäsuorat
u	18	vaadi	innostava	Kiev

fr-FR/fr-CA/fr-CH

Suprasegmentals for French

The Speech service phone set puts stress after the vowel of the stressed syllable. However, the **fr-FR** Speech service phone set doesn't support the IPA substress '̣'. If the IPA substress is needed, you should use the IPA directly.

Vowels for French

[Expand table](#)

sapi	ipa	viseme	Example 1	Example 2	Example 3
ae	a	2	arbre	patte	ira
af	ɑ	2		pâte	pas
an	ā	2	enfant	enfant	temps
ax	ə	1		petite	le
eh	ε	4	elle	perdu	était
eu	ø	1	œufs	creuser	queu
ey	e	4	ému	crétin	ôté
in	ē	4	important	peinture	matin
iy	i	6	idée	petite	ami
oe	œ	4	œuf	peur	
oh	ɔ	3	obstacle	corps	
on	ō	3	onze	rondeur	bon
ow	o	8	auditeur	beaucoup	pô
un	œ	4	un	lundi	brun
uw	u	7	outrage	introuvable	ou
uy	y	4	une	punir	élu

Consonant for French

[Expand table](#)

sapi	ipa	viseme	Example 1	Example 2	Example 3
b	b	21	bête	habille	robe
d	d	19	dire	rondeur	chaude
f	f	18	femme	suffixe	bof
g	g	20	gauche	égale	bague

sapi	ipa	viseme	Example 1	Example 2	Example 3
gn	ɲ	19			peigne
hw	ɥ	7	huile	nuire	
k	k	20	carte	écaille	bec
l	l	14	long	élire	bal
m	m	21	madame	aimer	pomme
n	n	19	nous	tenir	bonne
ng	ŋ	20			parking
p	p	21	patte	repas	cap
r	ʁ	13	rat	chariot	sentir
s	s	15	sourir	assez	passe
sh	ʃ	16	chanter	machine	poche
t	t	19	tête	ôter	net
v	v	18	vent	inventer	rêve
w	w	7	oui	fouine	
y	j	6	yod	piétiner	Marseille
z	z	15	zéro	raisonner	rose
	n̥	19			un arbre
	t̥	19			quand
	z̥	15			corps

1 Only for some foreign words.

💡 Tip

The fr-FR Speech service phone set doesn't support the following French liaisons, n̥, t̥, and z̥. If they are needed, you should consider using the IPA directly.

⚠ Note

fr-CA, fr-CH locales don't support SAPI phones now.

he-IL

Vowels for he-IL

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
i	6	איש	סיר	כלי
e	4	אש	כן	פה
a	2	אף	קו	מה
o	8	אות	יום	לא
u	7	עוגה	כרוב	הגיעו

Consonant for he-IL

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
p	21	פס	קליפה	טיפ
b	21	בד	סבא	פאב
t	19	טיפה	מתנה	חוט
d	19	דבר	אדום	תמיד
k	20	כתב	בוקר	חלק
g	20	גדול	אגף	דג
ʔ	19	אָבִיב	שיעור	
f	18	פילטר	סופר	סוף
v	18	וילון	כבד	לב
s	15	שמלה	כסף	פנס

ipa	viseme	Example 1	Example 2	Example 3
z	15	זאב	מזל	אגוז
ʃ	16	שולחן	פשוט	כביש
x	12	חתול	אוכל	פרח
h	12	הולך	זהב	בה
ʦs	19,15	צד	עצם	מומלץ
m	21	מאוד	סימן	חלום
n	19	נפש	תינוק	אבן
l	14	לשון	מילה	דגל
ʁ	13	ראשון	מורה	חיבור
j	6	ילד	מצוין	כדאי
ʒ	16	ז'אנר	מז'ור	'בז
tʃ	19,16	צ'יפ	קפוצ'ון	'סנדוויץ
dʒ	19,16	ג'ונגל	פיג'מה	'קוטג

hr-HR

Vowels for hr-HR

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
e	4	Egipat	Games	drveće
e:	4	eri	brijegom	de1taljne
i	6	ispada	želimo	javnosti
i:	6	iako	list	kompletni
u	7	ubacio	konkurentne	jednu
u:	7	Una	funta	Yu

ipa	viseme	Example 1	Example 2	Example 3
a	2	američke	kovačić	kredita
a:	2	anđela	gradila	Divulja
o	8	oaza	nanosi	do
o:	8	Olgu	kiselog	to

Consonant for hr-HR

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
d	19	dakle	evidenciji	kod
v	18	volja	građevinskog	imperativ
s	15	sabor	informisanja	interes
t	19	tad	informirati	ispit
n	19	na	žene	jeftin
l	14	logor	konstatirali	kapital
ʎ	14	ljudski	košulju	kralj
ʦ	19,15	carina	krivice	pravac
ʧ	19,16	četvorica	kritičan	osnivač
j	6	jednostavan	kuju	ovaj
x	12	hrvatskog	mahala	ovakvih
z	15	znanstvenom	mehanizacije	prijelaz
ʒ	16	žalbu	mladeži	crtež
r	13	red	moraju	dar
k	20	kažu	nakani	dnevnik
m	21	Mađara	napadima	dobrim
p	21	Poljska	napadnut	kamp

ipa	viseme	Example 1	Example 2	Example 3
g	20	gore	negativna	kaznenog
ʦ	16	ćelija	neisplaćene	mladić
f	18	fabula	nostrifikaciji	šef
b	21	Belgija	oba	sukob
d͡ʒ	19,16	džempera	Ilidže	George
ɲ	19	nje	emitiranja	stupanj
ʈ	16	đakovačkim	gađati	vođ
ʃ	16	šef	stišati	Glavaš

hu-HU

Vowels for hu-HU

Expand table

ipa	viseme	Example 1	Example 2	Example 3
ø	1	ördög	ördög	huszonkettő
øː	1	ők	önműködően	öntöző
a	2		Schumacher	
aː	2	árut	bizonyára	burzsua
ɛ	4	előzni	kisbéresnek	hogyne
eː	4	Édes	komédia	fölfelé
i	6	idegen	omnibuszok	kollégiumi
iː	6	ígyen	áhítat	
o	8	oldali	komor	Figo
o	2	atyját	olvasni	Olga
oː	8	ólmot	históriát	fénymásoló

ipa	viseme	Example 1	Example 2	Example 3
u	7	ugyanis	ezután	falu
u:	7	úrrá	fékút	számú
y	4	üdítőt	átsütve	alsóbbrendű
y:	4	űrallomás	gépjárművek	idejű

Consonant for hu-HU

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
b	21	busz	hiába	
b:	21		hasá bb urgonya	rövide bb
d	19	dévédé	fapados	szaporítanád
ɟ	16	gyártó	franciaágyas	huszonegy
d:	19		Goddess	hadd
ɟ:	16		ebédjén	hagyj
ɟ͡ʒ	19,16	Dzsó	menedzselési	college
ɟ͡ʒ:	19,16		Baggio	
dz	19,15	dzémsz	edző	McDonalds
dz:	19,15		eddeni	
f	18	figura	kofa	golf
f:	18		koffer	seriff
g	20	gondolom	számítógép	rúg
g:	20		faggatta	függ
h	12	hit	ruhában	ah
h:	12		ehhez	
j	6	János	olyan	karéj

ipa	viseme	Example 1	Example 2	Example 3
ɲ	19	nyakán	pecsenyét	példány
ʝ:	6	Ljeszkovai	majmolják	állj
ɲ:	19		pihenjen	
k	20	kofa	ruhákat	ruhák
k:	20		megkondult	makk
l	14	lent	rúla	evvel
l:	14		kolléga	áll
m	21	magyar	számít	órám
m:	21		anyámmal	kilogramm
n	19	népies	szótlantul	hiszen
ŋ	20		angel	
n:	19		onnan	fenn
p	21	Pál	tapogatódzó	számítógép
p:	21		beröppent	befejezéseképp
r	13	rág	óraára	órakor
r:	13		amerre	forr
s	15	számára	fölveszi	fölmész
ʃ	16	saját	förtelmesen	főorvos
s:	15		halasszuk	hazajössz
ʃ:	16		házassága	keress
t	19	Tata	rútak	hit
c	16		egyházmegye	dirty
t:	19		hitte	vágodott
c:	16		bátyja	Pretty
ts	19,15	címe	biciklis	huszonnyolc
tʃ	19,16	csigán	húgocskám	Gregorics

ipa	viseme	Example 1	Example 2	Example 3
ʦ:	19,15		játszad	játsz
ʦ̞:	19,16		barátságos	futballmeccs
v	18	varr	Olívia	Dönöv
v:	18		evvel	
x	12	hrabovszki	ihletével	
ɯ	20		alapján	Kapj
z	15	zúgást	csizmám	csimpánz
ʒ	16	zsûrivel	félmázsás	Balázs
z:	15		húzza	fékezz
ʒ:	16		garázssal	

id-ID

Vowels for id-ID

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
ə	1		benar	komite
a	2		babat	engga
ai̯	2,6	air	raikage	pantai
au̯	2,4	aurat	ataupun	pisau
e	4	energi	feri	tempe
ɛ	4	enrique	nenek	
i	6	indah	yakin	key
i	6	irama	biar	deflasi
ɔ	3	off	esok	law

ipa	viseme	Example 1	Example 2	Example 3
o	8	obat	bobot	domino
ɔɪ	3,6		reboisasi	sepoi-sepoi
u	7	umur	buah	linu
ʊ	4		duduk	

Consonant for id-ID

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
ʔ	19	amores	maaf	pencak
b	21	babat	nebak	dishub
d	19	deder	pedang	cloud
ɖʒ	19,16	jarum	penjajah	buruj
f	18	flat	aversi	genitif
g	20	gabus	dagel	hamburg
h	12	hama	tuhan	entah
ɲ	19	nyaman	menyuci	
j	6	yakin	jaya	baduy
k	20	ketan	aku	polsek
l	14	labu	talenta	vital
m	21	masuk	namanya	sinom
n	19	nada	sekunar	proton
ŋ	20	ngengat	kenanga	abang
p	21	pacar	hampa	cakap
r	13	rabu	dikurang	nasar
s	15	sabuk	tetesan	jenius

ipa	viseme	Example 1	Example 2	Example 3
ʃ	16	syarat	isyarat	british
t	19	tabir	adaptasi	durat
tʃ	19,16	cakap	dicari	
w	7	wajah	yuwana	
x	12	khusuk	akhirnya	barzakh
z	15	zakat	pezina	mahfuz

it-IT

Vowels for it-IT

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
a	2	amo	sano	scorta
ai	2,6	aics	abbaino	mai
au	2,7	audio	rauco	bau
e	4	eroico	venti	sapore
ɛ	4	elle	avvento	lacchè
ɛj	4,6	eira	email	lei
ɛu	4,7	euro	neuro	
ei	4,6		aseità	scultorei
eu	4,7	europeo	feudale	
i	6	italiano	vino	solì
u	7	unico	luna	zebù
o	8	obesità	straordinari	amico
ɔ	3	otto	botte	però

ipa	viseme	Example 1	Example 2	Example 3
ɔj	3,6		oppioidi	
oi	8,6	oibò	intellettualoide	Gameboy
ou	8,7		show	talkshow

Consonant for it-IT

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
b	21	bene	ebanista	Euroclub
b:	21		gobba	
tʃ	16	cenare	acido	french
tʃ:	19,16		braccio	
k:	20		pacco	Innsbruck
d	19	dente	adorare	interland
d:	19		cadde	
ʒ	15	zero	orzo	
ʒ:	15		mezzo	
f	18	fame	afa	alef
f:	18		beffa	bluff
ʤ	16	gente	agire	beige
ʤ:	16		oggi	
g	20	gara	alghe	smog
g:	20		fugga	Zuegg
ʎ	14	gli	ammiragli	
ʎ:	14		foggia	
ɲ:	19		bagno	

ipa	viseme	Example 1	Example 2	Example 3
ɲ	19	gnocco	padrigno	Montaigne
j	6	ieri	piede	freewifi
k	20	caro	anche	tic
l	14	lana	alato	col
l:	14		colla	full
m	21	mano	amare	Adam
m:	21		grammo	
n	19	naso	lana	non
n:	19		panna	
p	21	pane	epico	stop
p:	21		coppa	
r	19	rana	motore	per
r:	13		carro	Starr
s	15	sano	cascata	lapis
s:	15		cassa	cordless
ʃ	16	scemo	Gramsci	slash
ʃ:	16		ascia	fiches
t	19	tana	eterno	alt
t:	19		zitto	
ʈ	15	tsunami	turbolenza	subtests
ʈ:	15		bozza	
v	18	vento	avaro	Asimov
v:	18		bevvi	
w	7	uovo	duomo	Marlowe
z	15	smodato	casa	elections

ja-JP

The Speech service phone set for `ja-JP` is based on the native phone [Kana](#) set.

Please see the following tables for kana and corresponding viseme in parentheses.

Katakana for ja-JP

[Expand table](#)

Katakana	ア	イ	ウ	エ	オ
ア	ア (19,2)	イ (6,6)	ウ (7,6)	エ (19,4)	オ (19,8)
カ	カ (20,2)	キ (20,6)	ク (20,6)	ケ (20,4)	コ (20,8)
サ	サ (15,2)	シ (16,6)	ス (15,6)	セ (15,4)	ソ (15,8)
タ	タ (19,2)	チ (16,6)	ツ (19,15,6)	テ (19,4)	ト (19,8)
ナ	ナ (19,2)	ニ (19,6)	ヌ (19,6)	ネ (19,4)	ノ (19,8)
ハ	ハ (12,2)	ヒ (12,6)	フ (12,6)	ヘ (12,4)	ホ (12,8)
マ	マ (21,2)	ミ (21,6)	ム (21,6)	メ (21,4)	モ (21,8)
ヤ	ヤ (6,2)	n/a	ユ (6,6)	n/a	ヨ (6,8)
ラ	ラ (19,2)	リ (19,6)	ル (19,6)	レ (19,4)	ロ (19,8)
ワ	ワ (7,2)	n/a	n/a	n/a	ヲ (19,8)
	ン (19)	n/a	n/a	n/a	n/a

Katakana diacritics for ja-JP

[Expand table](#)

Katakana diacritics	ア	イ	ウ	エ	オ
ガ	ガ (20,2)	ギ (20,6)	グ (20,6)	ゲ (20,4)	ゴ (20,8)
ザ	ザ (15,2)	ジ (16,6)	ズ (15,6)	ゼ (15,4)	ゾ (15,8)
ダ	ダ (19,2)	ヂ (16,6)	ヅ (15,6)	デ (19,4)	ド (19,8)
バ	バ (21,2)	ビ (21,6)	ブ (21,6)	ベ (21,4)	ボ (21,8)

Katakana diacritics	ア	イ	ウ	エ	オ
パ	パ (21,2)	ピ (21,6)	プ (21,6)	ペ (21,4)	ポ (21,8)

Katakana Yōon for ja-JP

[Expand table](#)

Katakana Yōon	ヤ	ユ	ヨ
キ	キヤ(20,6,2)	キユ(20,6,6)	キヨ(20,6,8)
シ	シヤ(16,6,2)	シユ(16,6,6)	シヨ(16,6,8)
チ	チヤ(16,6,2)	チユ(16,6,6)	チヨ(16,6,8)
ニ	ニヤ(19,6,2)	ニユ(19,6,6)	ニヨ(19,6,8)
ヒ	ヒヤ(12,6,2)	ヒユ(12,6,6)	ヒヨ(12,6,8)
ミ	ミヤ(21,6,2)	ミユ(21,6,6)	ミヨ(21,6,8)
リ	リヤ(19,6,2)	リユ(19,6,6)	リヨ(19,6,8)
ギ	ギヤ(20,6,2)	ギユ(20,6,6)	ギヨ(20,6,8)
ジ	ジヤ(16,6,2)	ジユ(16,6,6)	ジヨ(16,6,8)
ヂ	ヂヤ(16,6,2)	ヂユ(16,6,6)	ヂヨ(16,6,8)
ビ	ビヤ(21,6,2)	ビユ(21,6,6)	ビヨ(21,6,8)
ピ	ピヤ(21,6,2)	ピユ(21,6,6)	ピヨ(21,6,8)

Examples for ja-JP

[Expand table](#)

Character	sapi	ipa
合成	ゴ'ウセ	go'wuseji
所有者	ショユ'ウ?ヤ	ɕjoju'wuɕja
最適化	サイテキカ+	sajitecika,

ko-KR

Vowels for ko-KR

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
a	2	아가씨	강	하다
ɛ	4	애국가	백	세번째
e	4	에너지	가겠구나	가게
ʊ	6	으레	으쓱해	가스
i	6	이거	아직까지	어미
ʌ	1	어미	차선변경	가시겠어
o	8	오래	의혹을	차고
u	7	우간다	은둔자의	가시나무
ɰi	20,6	의자	배추흰나비가	가요계의
ø	1	외가댁	배치된다	하되
ʷa	7,2	와글와글	가치관과	가지와
ʷɛ	7,4	왜관	호쾌가	안돼
ʷe	7,4	웨딩드레스		
ʷi	7,6	위계적	구조위원회가	한가위
ʷʌ	7,1	워낙	가까워서	가까워
ja	6,2	야구	기술집약도가	끌려가야
jɛ	6,4	애가	가수애기예요	
je	6,4	예감	적대관계가	의례
jʌ	6,1	여가	감정평가사	이리하여
jo	6,8	요구가	사용중지	가거든요
ju	6,7	유가적	경제교류가	소유

Consonant for ko-KR

ipa	viseme	Example 1	Example 2	Example 3
b	21	바가		밥
p	21	빠른가	막부정부가	
b	21		사범학교가	
t͡ɕ^h	19,16	참가비는	아침과	
d̚	19	동네가		바깥
t	19	따라가지	깍두기가	
d	19		산도가	
ɡ	20	가쵸		각
k	20	까마귀가	젓가락으로	
g	20		단추가	
h	12		손가락질하거나	
ɦ	12	하기가	손가락질하며	
d͡ʑ	19,16		손잡이가	
d͡ʑ̚	19,16	자유가		
t͡ɕ	19,16	짜가면서	손가락질할	
k^h	20	키가	아킬레스건	
l	14			국가체제를
m	21	마다가스카르	통나무가	기침
n	19	나가서는	아느냐	따라가다보면
ŋ	20		강아지	한강
p^h	21	파티가	아파트	
r	19	라디오가	아름답게	
s^h	15	사고가	아스팔트	
s	15	쌍둥이가	멕시코가	

ipa	viseme	Example 1	Example 2	Example 3
t ^h	19	택시가	여타의	

ms-MY

Vowels for ms-MY

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
i	6	ibu	iklim	ahli
u	7	uang	buah	bahu
ə	1		kerja	nasionalisme
e	4	edar	aktres	kue
o	8	orang	anggota	pidato
a	2	anjing	anak	ada
ai	2,6			ceraai
au	2,7	auto	akaun	bakau
oi	8,6			amboi

Consonant for ms-MY

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
p	21	pekat	sekeping	cakap
b	21	banjir	lebih	jawab
t	19	tidak	peta	sikit
d	19	dekat	adakah	akad
k	20	ketat	akak	book

ipa	viseme	Example 1	Example 2	Example 3
g	20	gabung	bogel	dialog
ʔ	19	abang	kepercayaan	letak
tʃ	19,16	cepat	baca	
dʒ	19,16	jabatan	aja	kolej
m	21	memang	laman	malam
n	19	negeri	tanam	taman
ɲ	19	nyanyi	tanya	
ŋ	20		mangga	sayang
f	18	filem	artifak	aktif
v	18	vaksin	aktiviti	
s	15	sahabat	akses	tumis
z	15	zaman	lazat	
ʃ	16	syarikat	bersyarat	
x	12	khabar	akhir	tarikh
r	13	racun	merah	lebar
h	12	hingga	aduhai	boleh
j	6	yang	ayah	
w	7	walaupun	bawah	
l	14	lidah	alam	katil

nb-NO

Vowels for nb-NO

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
ɑ	2	annonse	betrakte	hoppa
æ	1	ergre	Palermo	
æ:	1	ærlig	belære	bæ
ɑ:	2	are	betale	bedra
ɛ	4	energi	kadetten	hoppe
ø:	1	øre	behøve	adjø
e:	4	ener	berede	distre
ɪ	6	ikke	setningen	taxi
i:	6	Eagle	bevise	konditori
ɔ	3	åtte	kontrollen	altså
œ	4	ønske	belønning	Mossø
o:	8	år	område	begå
u	7	okse	økokrim	ego
u:	7	ord	telefonen	bidro
ʏ	7	ytterst	benytte	Ally
ʊ	6	under	forundret	jaggu
ʊ:	6	ule	umulig	intervju
y:	4	yte	belyse	paraply
æɪ	1,6	eiendom**	utleide	snarvei
æʊ	1,6	aura	Litauen	fortau
ɑɪ	2,6	aibel**	Aserbajdsjan	Dubai
œʏ	4,7	øyer	ablegøyer	syltetøy
ɔʏ	3,7	Oilers	boikotten	konvoi
ʊɪ	6,6		Bruins	Mitsui

Consonant for nb-NO

ipa	viseme	Example 1	Example 2	Example 3
p	21	pil	ape	lapp
t	19	tall	matte	matt
k	20	kall	jakke	takk
b	21	bil	klubbe	lobb
d	19	dal	lide	gadd
g	20	gås	sagen	lag
f	18	fil	klaffe	klaff
h	12	hall	beholde	
s	15	sil	vise	viss
ʃ	15	sju	maskin	dusj
ç	12	tjukk	bekjenne	Korch
v	18	vår	leve	lov
m	21	mil	komme	lam
n	19	nål	minnes	søvn
ŋ	20		penger	lang
l	14	løs	måle	tal
r	13	ris	karre	tørr
j	6	jag	utjevne	detalj
ɖ	19		burde	ferd
ʎ	14		farlig	jarl
ɳ	19		barnet	jern
t	19		skjorte	gjort

nl-NL/nl-BE

Vowels for nl-NL

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
ɑ	2	af	bak	bavarois
a:	2	aan	maal	ja
ā	2	enfin	manchet	croissant
au	2,7	oud	bout	hou
ɛ	4	en	weg	hè
e:	4	één	heet	nee
ɛ:	4	airbag	blèr	
ɛi	4,6	eis	wijn	zij
ē	4		lingerie	elektricien
ø:	1	euro	deur	milieu
ɪ	6	ik	ding	
i	6	iets	sliep	drie
ɔ	3	op	slot	joh
u	7	oefen	hoed	doe
ɔ:	3		roze	
ō	3			Macron
o:	8	ook	boom	zo
ʏ	7	urn	dus	
ə	1	een	trommel	de
œy	4,4	uil	juist	bui
œ	4	oeuvre	service	
y	4	uur	tuur	nu

Consonant for nl-NL

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
b	21	boos	fobie	
d	19	dat	kudde	
f	18	fiets	gefeest	blaf
χ	12	ga	magie	hoog
ʔ	19		beamen	
ɦ	12	hoek	behaard	
g	20	gujarati	again	drug
j	6	jij	boeien	haai
k	20	kat	haken	zwak
l	14	land	familie	kool
m	21	man	demon	raam
n	19	niks	kannon	pan
ŋ	20		brenge	zing
p	21	poer	rapen	heb
ʀ	13	romp	waarom	kier
s	15	soms	precies	heus
ʃ	16	sjaal	vaasje	lunch
t	19	tot	laten	groot
w	7		flauwe	follow
v	18	voor	haven	
u	18	wat	fusiewet	
z	15	zal	lezen	
ʒ	16	jus	beige	

pl-PL

Vowels for pl-PL

[Expand table](#)

ipa	viseme	Example 1	Example 2	Example 3
a	2	autor	kolację	karasia
ɛ	4	elbląska	reflektor	osiągnięcie
ɛ̃	4		nieczęsto	skorupę
i	6	informatycznym	powiśle	gadali
ĩ	6		cudzymi	rodziły
ɔ	3	osobniki	uboju	rogatko
ɔ̃	3		mąż	intelektualistą
u	7	unosimy	arkuszy	przeznaczeniu

Consonant for pl-PL

[Expand table](#)

ipa	viseme	Example 1	Example 2	Example 3
b	21	bliska	wyśrubowane	
bʲ	21	biuro	zapobiegać	
t͡ɕ	19,16	ciągników	uczuciowych	suchość
t͡ɕ̞	19,15	częstkowe	odpoczynek	niszcz
c	16	kije	lisikiewicz	
d	19	drogowy	porodówkę	bastard
dʲ	19	dializy	studiuję	
d͡z	19,15	dzwonnica	wrodzony	
d͡z̞	19,16	dziurawe	pochodziłam	

ipa	viseme	Example 1	Example 2	Example 3
f	18	wprost	długofalowy	konserwantów
f ^j	18	filmoteki	grafiką	
g	20	geometrią	nawigacja	
ʒ	16	gitarzystą	religijnym	
ɖʒ	19,15	dżunglę	menedżerskie	
k	20	królestwa	naukowo	matematyk
l	14	latający	populacje	handel
l ^j	14	lisek	okolica	
m	21	majątkowy	wytłumaczenia	błagam
m ^j	21	mieszkającej	dynamicznie	
n	19	napędu	poczynaniach	balkon
ŋ	20		ciągłość	
ɲ	19	niewidoczna	zmieniała	poznań
p	21	potoczne	terapeuci	odstęp
p ^j	21	pijawek	skupieniu	
r	13	regionu	operową	administrator
r ^j	13	ripostuje	imperialnej	
s	15	solone	przetrasowania	biogaz
ɕ	16	sierpień	doniesieniem	mogłaś
ʃ	16	szanowanych	wpatrzeniu	skręcasz
t	19	talentom	kwaterze	dowód
t ^j	19	tirami	marketingiem	
ʈʂ	19,15	cyfrą	agencyjne	pałac
v	18	wysłaniu	przeprowadzają	
v ^j	18	winiarstwa	bukowianka	
w	7	łączenie	prałata	drukował

ipa	viseme	Example 1	Example 2	Example 3
x	12	hamulcem	zdychają	kostiumach
x ^j	12	hiszpańscy	psychice	
j	6	jeździła	popijałam	najważniejszej
z	15	zaskakują	partyzanckich	wiz
z	16	ziemniaki	zgryzienia	
ʒ	16	żyrandol	nowożytniej	

pt-BR

Vowels for pt-BR

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
i	6	ilha	ficar	comi
ĩ	6	intacto	pintar	aberdeen
a	2	água	dada	má
ɔ	3	ora	porta	cipó
u	7	ufanista	mula	peru
ũ	7	uns	pungente	kuhn
o	8	ortopedista	fofo	avô
e	4	elefante	elefante	você
ẽ	4	anta	canta	amanhã
ə	1	aqui	amaciar	dada
ɛ	4	ela	serra	até
ē	4	endorfina	pender	
õ	8	ontologia	conto	

Consonant for pt-BR

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
ɰ	7			atualização
w	7	washington	água	usou
p	21	pato	capital	
b	21	bola	cabeça	
t	19	tato	rato	
d	19	dado	amado	
g	20	gato	maragato	
m	21	mato	comer	
n	19	no	ano	
ɲ	19	nhoque	ninho	
f	18	faca	afago	
v	18	vaca	cavar	
r	19		para	amar
s	15	satisfeito	amassado	casados
z	15	zebra	azar	
ʃ	16	cheirar	machado	
ʒ	16	jaca	injusta	
x	12	rota	carreta	
tʃ	19,16	tirar	atirar	
dʒ	19,16	dia	adiar	
l	14	lata	aleto	
ʎ	14	lhama	malhado	
ɣ	6		inabalavelmente	hífen

ipa	viseme	Example 1	Example 2	Example 3
j	6		caixa	sai
k	20	casa	ensacado	

pt-PT

Vowels for pt-PT

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
a	2	ábdito	consular	medirá
e	4	abacaxi	domação	longa
ej	4,6	eidético	direita	detectei
ẽ	4	anverso	viajante	afã
ẽj	4,6	angels	viagens	também
ẽw	4,7	hãõ	significaçãozinha	gabão
ew	4,7		saudar	hello
aj	2,6	airosa	culturais	vai
ɔ	3	hora	depósito	ló
ɔj	3,6	óis	heróico	dói
aw	2,7	outlook	incauto	pau
ə	1	extremo	sapremar	noite
e	4	eclipse	haver	buffet
ɛ	4	eco	hibérnios	paté
ɛw	4,7		pirinéus	escarcéu
ẽ	4	embaçado	dirimente	ámen
ew	4,7	eu	deus	bebeu

ipa	viseme	Example 1	Example 2	Example 3
i	6	igreja	aplaudido	escrevi
ĩ	6	impaciente	espinçar	manequim
ɪw	6,7		niue	garantiu
o	8	ofir	consumidor	stacatto
oɟ	8,6	oirar	noite	foi
õ	8	ombrão	barronda	dom
õɟ	8,6		ocupações	expõe
u	7	ubi	facultativo	fado
uɟ	7,6	uivar	arruivado	fui
ũ	7	umbilical	funcionar	fórum
ũɟ	7,6		muito	

Consonant for pt-PT

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
b	21	bacalhau	tabaco	club
d	19	dado	dado	band
r	19	rename	verás	chutar
f	18	fim	eficácia	golf
g	20	gadinho	apego	blog
j	6	iode	desassociado	substitui
k	20	kiwi	traficado	snack
l	14	laborar	pelada	full
ɭ	14		polvo	brasil
ʎ	14	lhanamente	antilhas	

ipa	viseme	Example 1	Example 2	Example 3
m	21	maça	amanhã	modem
n	19	nutritivo	campana	scan
ɲ	19	nhambu-grande	toalhinha	penh
p	21	pai	crápula	laptop
ʀ	13	recordar	guerra	chauffeur
s	15	seco	grosseira	boss
ʃ	16	chuva	duchar	médios
t	19	tabaco	pelota	input
v	18	vaca	combatível	pavlov
w	7	waffle	restituir	katofio
z	15	zâmbia	prazer	jazz
ʒ	16	gelada	infligir	cuj

ro-RO

Vowels for ro-RO

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
ə	1	ăsta	abătut	fizică
ɨ	6	înspre	hotărâre	îhî
a	2	absolut	prematur	Praga
e	4	educație	tablete	alianțe
ɛa	4,2		studentească	badea
eo	4,8		bleumarín	vreo
i	6	Italia	aripi	aberații

ipa	viseme	Example 1	Example 2	Example 3
o	8	oricum	catacombe	radio
ɔa	8,2	oară	închisoare	șamoa
u	7	umble	gradul	Alexandru

Consonant for ro-RO

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
b	21	băi	vizibil	arab
b ^j	21			microbi
d	19	doctor	video	miliard
ɖʒ	19,16	ger	vegetație	
ɖʒ ^j	19,16			colegi
f	18	furtună	efort	ecograf
f ^j	18			filosofi
g	20	galeria	egal	filolog
g ^j	20		neghină	unghi
h	12	hexagon	arhitect	monarh
j	6	ieri	băiat	ditai
k	20	cadru	baricadat	pitic
k ^j	20			urechi
l	14	laptop	alint	Paul
l ^j	14			circuli
m	21	mandarine	camera	atom
m ^j	21			consumi
n	19	nepot	Canada	Eden

ipa	viseme	Example 1	Example 2	Example 3
ŋ	20		banca	plâng
n ^j	19			dragoni
p	21	pai	opera	Filip
p ^j	21			ocupi
r	13	real	mere	distribuitor
r ^j	13			palmieri
s	15	sertar	căsătorit	exclus
ʃ	16	șine	cușetă	greș
ʃ ^j	16			groși
t	19	teracota	material	abonament
t ^j	19			foști
ts	19,15	țar	cuțit	vorbăreț
tʃ	19,16	circa	meciuri	
ts ^j	19,15			uscați
tʃ ^j	19,16			indici
v	18	vaccin	gravidă	fugitiv
v ^j	18			nervi
w	7	uau	cuantificare	pliu
z	15	zoologică	frază	parbriz
ʒ	16	jar	abajur	pasaj
z ^j	15			semnezi
ʒ ^j	16			dârji

ru-RU

Vowels for ru-RU

[Expand table](#)

ipa	viseme	Example 1	Example 2	Example 3
a	2	адрес	радость	беда
л	1	облаков	застенчивость	внучка
ə	1		яблочного	
ε	4	эпос	белка	кафе
i	6	иней	лист	соловьи
ɪ	6	игра	медведь	мгновение
ɨ	6	энергия	лысый	весы
ɔ	3	окрик	мот	весло
u	7	ужин	куст	пойду

Consonant for ru-RU

[Expand table](#)

ipa	viseme	Example 1	Example 2	Example 3
p	21	профессор	поплавок	укроп
pʲ	21	Петербург	ослепительно	степь
b	21	большой	собака	
bʲ	21	белый	убедить	
t	19	тайна	старенький	твид
tʲ	19	тепло	учитель	синеть
d	19	доверчиво	недалеко	
dʲ	19	дядя	единица	
k	20	крыло	кукуруза	кустарник
kʲ	20	кипяток	неяркий	
g	20	гроза	немного	

ipa	viseme	Example 1	Example 2	Example 3
g ^j	20	герань	помогите	
x	12	хороший	поход	дух
x ^j	12	хилый	хихиканье	
f	18	фантазия	шкафах	кров
f ^j	18	фестиваль	кофе	верфь
v	18	внучка	синева	
v ^j	18	вертеть	свет	
s	15	сказочник	лесной	карапуз
s ^j	15	сеять	посередине	зажглись
z	15	заяц	звезда	
z ^j	15	земляника	созерцал	
ʃ	15	шуметь	пшено	мышь
ʒ	15	жилище	кружевной	
t͡s	19,15	целитель	Венеция	незнакомец
t͡ʃ	19,16	часы	очарование	мяч
ʧ:	16	щелчок	ощущать	лещ
m	21	молодежь	несмотря	том
m ^j	21	меч	дымить	семь
n	19	начало	оконце	сон
n ^j	19	небо	линялый	тюлень
l	14	лужа	долгожитель	мел
l ^j	14	лицо	недалеко	соль
r	13	радость	сорока	двор
r ^j	13	рябина	набережная	дверь
j	6	есть	маяк	игрушечный

sk-SK

Vowels for sk-SK

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
i	6	idea	efektivita	éry
e	4	edícia	absencia	farme
a	2	abnormálne	eskapáda	ária
o	8	oba	banková	esperanto
u	7	udial	február	babu
ɤ	6		mäsité	
i:	6	ílu	edícii	druhý
e:	4	éra	anamnézy	ázijské
a:	2	áno	animácia	druhová
o:	8	ódy	bilión	haló
u:	7	úbočí	absolútna	druhú
ɪa	6,2		piatkové	maškrtia
ɪe	6,4		domnienka	námestie
ɪu	6,7			väčšiu
uo	7,8	ôsma	jahôd	malinô
au	2,7	audio	aplaudovalo	sredau
ou	8,7			
ə	1			

Consonant for sk-SK

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
p	21	pád	apelu	cap
b	21	babička	abiogenézy	dub
t	19	tabak	foto	art
d	19	delta	editor	backhand
c	16	ťahač	docenti	farnost'
ʃ	16	d'alej	strede	lod'
k	20	kabaret	bábkový	chudáčik
g	20	galón	demagógia	mozog
ʦ	19,15	certifikácie	cicavce	bác
ʤ	19,15		medzí	
tʃ	19,16	čajový	frčí	boháč
dʒ	19,16	džúsu	brandže	
f	18	fabrík	biografie	fotograf
v	18	vokály	evanjelické	
s	15	sekunda	esá	algoritmus
z	15	zábal	fazuľu	aníz
ʃ	16	šabl'a	duševná	chápeš
ʒ	16	žaba	veži	kaluž
x	12	charakter	biochémie	bohatých
h	12	háčik	bohmi	
r	13	rabat	eróziou	éter
ɾ	13		chatrče	leicester
ɾː	13		vrtal	
l	14	lampa	električka	čakal
ɫ	14		dĺhý	nóbl
ɫː	14		jabl'k	

ipa	viseme	Example 1	Example 2	Example 3
ʎ	14	l'ad	citel'né	byl'
m	21	meter	emisný	aktom
ɱ	21		amfiteáter	
n	19	nábeh	colnému	fajn
ɳ	19		slovinský	
ŋ	20		banket	
ɲ	19	ňom	ani	jačmeň
ɥ	7		cestovní	aktív
ɨ	6		fajka	chatovej
j	6	ja	eseje	
w	7	vzbudí	krivdí	

sl-SI

Vowels for sl-SI

Expand table

ipa	viseme	Example 1	Example 2	Example 3
ə	1	rjava	pisemski	december
a	2	azilantov	hišam	delniška
a:	2	avto	marketplace	lucia
ɛ	4	edicija	pridem	ničle
e:	4	ebola	pridevnik	prepove
ɛ:	4	ena	producenta	janže
i	6	ideja	poudarim	poudariti
i:	6	igla	ilirska	jedmi

ipa	viseme	Example 1	Example 2	Example 3
ɔ	3	oba	morfološke	Marko
ɔː	3	oče	črnomorskem	
oː	8	občina	reformam	seno
u	7	ulova	mamut	mandatu
uː	7	ura	dramaturgom	intervju

Consonant for sl-SI

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
b	21	brez	bober	lebdeča
d	19	dajal	degradacija	navedbah
d ¹	19	dleto		navedla
dn	19,19	dna		dohodne
ďž	19,16	džezovske	bridža	menedžment
ďž	19,15	odzivajo		Kocbek
f	18	fagota	fotografa	golf
ɱ	21			nimfa
ɣ	20			ahdeloja
g	20	gaber	lagal	ragbi
ɪ	6		pojdi	emisij
j	6	jadra	kajak	najostreje
k	20	kabel	akademija	alkoholik
l	14	labirint	olajša	šal
l ^j	14			poljski
m	21	maček	omara	potem

ipa	viseme	Example 1	Example 2	Example 3
ŋ	20	Ngaliemski_slapovi		banka
n	19	nabave	obarvana	obarvan
nʲ	19			konjska
p	21	pada	sapa	sestop
r	13	rabila	sorazmerna	spor
s	15	srajca	virusa	virus
ʃ	16	šah	sušijo	tovariš
t	19	tabela	gojiti	flavtist
t¹	19	tla	škatla	desetletne
tn	19,19			devetnajst
tʃ	19,16	čaj	gneča	hlač
ts	19,15	car	raca	rejec
u	7			avto
v	18	veja	avanture	
w	7	vgradila		slabokrvnost
m	7	vstati		
x	12	hiša	jahači	jamah
ʒ	16	žaba	ježa	možgani
z	15	zmaj	doza	izvoza

sv-SE

Vowels for sv-SE

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
a	2	andas	betrakta	gryta
æ	1	ärter	smärta	
æ:	1	ärliga	besvära	
ɑ:	2	avig	betala	bedra
ɔ	3	åtta	kontrollen	jaså
a_u	2,7	aura	pauser	Olau
ə	1		äpplena	pojke
e	4	energi	servetten	Arjepluovve
ɛ	4	äpple	berätta	Siviä
ɛ:	4	äta	beträda	trä
e:	4	eka	konkreta	café
œ	8	ört	störta	
œ:	4	ören	beröra	
œ	4	öppen	Alströmer	Päiviö
ø:	1	öl	belöning	adjö
ɪ	6	idé	vitsippa	kiwi
i:	6	ivrig	kris	parti
ʊ	4	oas	betrodda	konto
u:	7	oro	förtroende	bero
o:	8	åtala	telefonen	nivå
ə	1	uppenbar	förundrad	farstu
ʊ:	6	ute	bestulen	intervju
y	4	ytterst	rykte	Tommy
y:	4	yta	förtydliga	paraply

Consonant for sv-SE

ipa	viseme	Example 1	Example 2	Example 3
p	21	pil	apa	topp
t	19	tal	matta	akut
k	20	kål	jacka	tak
b	21	bil	klubba	jobb
d	19	dal	lida	stad
g	20	gås	såga	lag
f	18	fil	klaffa	klaff
h	12	hal	behålla	
s	15	sil	visa	viss
ʃ	16	sjuk	maskin	dusch
ç	16	tjock	åkkänslan	coach
v	18	vår	leva	torv
m	21	mil	kamma	arm
n	19	nål	minnas	sömn
ŋ	20		ringa	ung
l	14	lös	måla	tal
r	13	ris	kärra	borr
j	6	jag	tröja	haj
ɖ	19		borda	bord
ʎ	14		porlande	kärl
ɳ	19		gärna	barn
ʂ	15		forsa	fors
ʈ	19		skjorta	gjort

Vowels for th-TH

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
a	2		นะ	
a:	2		หาด	หา
e	4		เตะ	
e:	4		เท	ได้
i	6		ปิด	
i:	6		ปีก	ปี
ia	6,2		เจียบ	
o	8		บด	
o:	8		แลกโตส	ฉาวโฉ่
ə	1		ใบฝักเงิน	
ə:	1		เกิน	เหม่อ
u	7		จุด	
u:	7		ดูด	ดู
ua	7,2		กวน	รัว
ʊ	6		ยัด	
ʊ:	6		มิด	มีอ
ʊa	6,2		เรื่อน	เรื่อ
ɛ	4		คู่แข่ง	
ɛ:	4		แบน	คำแปล
ɔ	3		น็อค	
ɔ:	3		นอน	เจ็บคอ

Consonant for th-TH

[Expand table](#)

ipa	viseme	Example 1	Example 2	Example 3
b	21	บิน	กระบี่	ยมน
t̚	19,16	เจ็ด	กระโจม	
t̚ʰ	19,16	เข้าๆ	วันอาสาฬหบูชา	
d	19	เดช	วิดีโอ	
f	18	ฟ้า	เฟื่องฟู	
h	12	หา	เศษอาหาร	
j	6	ยา	เรียดนด	ยาย
k	20	กา	เนื้อไก่	มาก
kʰ	20	คา	เลขา	
l	14	ลา	หูลลาม	
m	21	มา	เสมอ	รัดกุม
n	19	นา	แอมโมเนีย	กาล
ŋ	20	งา	แต่งงาน	แหล่ง
p	21	ปา	โดยทั่วไปแล้ว	กับ
pʰ	21	พา	ลัภัย	
r	13	รา	ไข่มาลาเรีย	
s	15	สาม	ไม่มีสติ	
t	19	ตา	กตিকা	อาจ
tʰ	19	ทา	คุณธรรม	
w	7	วัง	กวี	แก้ว
ʔ	19	อา	สะอาด	

Tone for th-TH

[Expand table](#)

ipa	Description	Example
˥	Mid	ไป bpai ^M (to go)
˨	Low	ไข่ khai ^L (egg)
˥˩	Falling	ใช่ chai ^F (yes; agreement)
˥˥	High	ครับ khrap ^H ([spoken by a male] yes)
˨˨˩	Rising	หนัง nang ^R (cinema film)

tr-TR

Vowels for tr-TR

[Expand table](#)

ipa	viseme	Example 1	Example 2	Example 3
a	2	armut	fal	elma
ɑː	2	ağrı	kağrı	dağ
e	4	erik	kel	akide
eː	4	eğri	değnek	yeğ
œ	4	ördek	göl	banliyö
œ̞	4,16	öğlen	açıköğretim	
i	6	ilaç	kil	kedi
i̞	6,16	iğne	mevkiine	tebliğ
o	8	orman	kol	vazo
o̞	8,16	oğlan	doğru	doğ
u	7	uçak	kuş	koku
u̞	7,16	uğur	buğra	başbuğ
ʊ	6	ıhlamur	tıp	kazı
ʊ̞	6,16	ığdır	sıgılık	tığ

ipa	viseme	Example 1	Example 2	Example 3
y	4	ülke	gül	öykü
ʏ̯	4,16		düğme	

Consonant for tr-TR

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
b	21	balık	ebe	sertab
c	16	keçi	ekşi	bölük
tʃ	19,16	çöp	uçak	ilgeç
d	19	dere	badem	ad
f	18	fil	kefen	çarşaf
g	20	gaz	yorgun	diyalog
ɣ	20		ağır	
ʒ	16	gavur	kevgir	
h	12	halat	ahır	külâh
j	6	yer	ayva	olay
dʒ	19,16	cezve	evcimen	hac
k	20	kabak	bakla	çocuk
l	14	leylek	delik	kil
ʙ	6	lala	kalın	pul
m	21	muz	yemek	kalem
n	19	nar	inek	sorun
ŋ	20		mangal	ring
p	21	para	kapı	rakip
r	19	renk	iri	minder

ipa	viseme	Example 1	Example 2	Example 3
s	15	sal	kısa	bahis
ʃ	16	şekil	koşu	afiş
t	19	terlik	kutup	adalet
v	18	vadi	tava	ev
w	7		tavuk	
z	15	zemin	gezi	kaz
ʒ	16	jöle	angajman	refüj

vi-VN

Vowels for vi-VN

 Expand table

ipa	viseme	Example 1	Example 2	Example 3
a	2		ban	ma
ɛ	4		hẹp	mẹ
i	6			chí
ɔ	3		học	tò
u	7		hung	thù
ua	7,2			chúa
aj	2,6			tài
ɛj	4,6			lấy
əj	1,6			chơi
o	8			xô
iew	6,4,7			riêu
ɪə	6,1		khuyết	

ipa	viseme	Example 1	Example 2	Example 3
ɔ̌i	3,6			mọi
ə	1		phần	
ie	6,4		biển	
ǔj	7,6			mùi
āw	2,7			bảo
ɪ	6		chứng	tử
e	4		nhặt	
ǎw	2,7			màu
ǎj	2,6			ngày
ɪə̌j	6,1,6			lưới
ǒj	8,6			hồi
ə:	1		lợn	sở
e	4		sên	tế
ɔ̌w	3,7			tấu
ēw	4,7			béo
l̄w	6,7			tịu
ɪ̄w	6,7			tựu
ěj	4,6			bêu
ɪ̄^w	6,1,7			bươu
ɪ̄j	6,6			củi
ɪ	6	inh	tinh	
iə	6,1			kìa
ā̌j	2	ach	họach	

Consonant for vi-VN

ipa	viseme	Example 1	Example 2	Example 3
b	21	ba		
k	20	canh		diệc
z	15	diễn		
j	6	giặc		
ɟ	13	róc		
f	18	phụ		
ɣ	20	gà		
h	12	hoa		
l	14	làm		
m	21	mười		nắm
n	19	núi		tiền
p	21	pin		pháp
s	15	xào		
ʃ	15	son		
t	19	tuổi		hết
v	18	vàng		
ɖ	19	đón		
ŋ	20	ngủn		lưng
x	12	khùng		
ɲ	19	nhàm		sanh
tʰ	19	thông		
ʈ	19	trùng		
tʃ	19,16	chim		chỉ
w	7		hoang	

Tone for vi-VN

[Expand table](#)

ipa	Description	Example
˧	tone ngang	xem
˨˨˩	tone huyền	làm
˨˩˩	tone sắc	sống
˨˩˨˩	tone hỏi	phải
˨˩˨˩˩	tone ngã	cũng
˨˩˩˩	tone nặng	một

zh-CN

The Speech service phone set for `zh-CN` is based on the native phone [Pinyin](#) set.

Pinyin Initials for zh-CN

[Expand table](#)

Pinyin	viseme	Character example	sapi example
b	21	玻	bo 1
p	21	坡	po 1
m	21	摸	mo 1
f	18	佛	fo 2
d	19	得	de 2
t	19	特	te 4
n	19	呢	ne 5
l	14	乐	le 4
g	20	哥	ge 1
k	20	科	ke 1

Pinyin	viseme	Character example	sapi example
h	12	喝	he 1
j	16	基	ji 1
q	16	欺	qi 1
x	16	希	xi 1
zh	19, 15	知	zhi 1
ch	19, 15	吃	chi 1
sh	15	诗	shi 1
r	15	日	ri 4
z	15	资	zi 1
c	15	此	ci 3
s	15	思	si 1
y	6	衣	yi 1
w	7	屋	wu 1

Pinyin Finals for zh-CN

 Expand table

Pinyin	viseme	Character example	sapi example
a	2, 2 (19, 2, 2 for no initials)	法	fa 3
o	7, 8, 8 (19, 8, 8 for no initials)	泼	po 1
e	1, 1 (19, 1, 1 for no initials)	歌	ge 1
i	6, 6, 6	理	li 3
u	7, 7, 7	步	bu 4
v	7, 4, 4	女	nv 3
ai	2, 4 (19, 2, 4 for no initials)	百	bai 3
ei	4, 6 (19, 4, 6 for no initials)	北	bei 3
ui	7, 4, 6	对	dui 4

Pinyin	viseme	Character example	sapi example
ao	2, 8 (19, 2, 8 for no initials)	号	hao 4
ou	8, 7 (19, 8, 7 for no initials)	走	zou 3
iu	6, 8, 7	牛	niu 2
ie	6, 4, 4	谢	xie 4
ue	7, 4, 4	略	lue 4
er	19, 1, 1	耳	er 3
an	2, 19 (19, 2, 19 for no initials)	喊	han 3
en	1, 19 (19, 1, 19 for no initials)	肯	ken 3
in	6, 6, 19	宾	bin 1
un	7, 1, 19	尊	zun 1
ang	2, 20 (19, 2, 20 for no initials)	朗	lang 3
eng	1, 20	恒	heng 2
ing	6, 1, 20	赢	ying 2
ong	7, 7, 20	红	hong 2
ia	6, 2, 2	家	jia 1
ian	6, 2, 19	面	mian 4
iang	6, 2, 20	象	xiang 4
iao	6, 2, 8	表	biao 3
iong	7, 7, 20	兄	xiong 1
ua	7, 2, 2	花	hua 1
uai	7, 2, 4	帅	shuai 4
uan	7, 2, 19	短	duan 3
uang	7, 2, 20	广	guang 3
uo	7, 8, 8	多	duo 1

Pinyin Whole syllable for zh-CN

Pinyin	viseme	Character example	sapi example
zhi	19, 15, 6, 6	知	zhi 1
chi	19, 15, 6, 6	吃	chi 1
shi	15, 6, 6	诗	shi 1
ri	15, 6, 6	日	ri 1
zi	15, 6, 6	资	zi 1
ci	15, 6, 6	词	ci 2
si	15, 6, 6	斯	si 1
yi	6, 6, 6	衣	yi 1
wu	7, 7, 7	屋	wu 1
yu	7, 4, 4	鱼	yu 2
ye	6, 4, 4	叶	ye 4
yue	7, 4, 4	月	yue 4
yuan	7, 2, 19	圆	yuan 2
yin	6, 6, 19	音	yin 1
yun	7, 1, 19	云	yun 1
ying	6, 1, 20	英	ying 1
a	19, 2, 2	啊	a 1
o	19, 8, 8	噢	o 1
e	19, 1, 1	鹅	e 2
ai	19, 2, 4	爱	ai 4
ei	19, 4, 6	欸	ei 1
ao	19, 2, 8	奥	ao 4
ou	19, 8, 7	偶	ou 3
an	19, 2, 19	安	an 1
en	19, 1, 19	恩	en 1

Pinyin	viseme	Character example	sapi example
ang	19, 2, 20	昂	ang 2

Pinyin Tone for zh-CN

[Expand table](#)

Pinyin	Character example	sapi example
bā	八	ba 1
bá	拔	ba 2
bǎ	把	ba 3
bà	坝	ba 4
ba	吧	ba 5

Example for zh-CN

[Expand table](#)

Character	Speech service
组织关系	zu 3 - zhi 1 - guan 1 - xi 5
累进	lei 3 - jin 4
西宅巷	xi 1 - zhai 2 - xiang 4
一会儿	yi 2 - hui r 4

zh-HK

The Speech service phone set for **zh-HK** is based on the native phone [Jyutping](#) [↗](#) set.

Jyutping Initials for zh-HK

[Expand table](#)

Jyutping	Character example	sapi example	VisemeID
p	怕	paa 3	21
b	巴	baa 1	21
t	他	taa 1	19
d	打	daa 2	19
k	卡	kaa 1	20
g	家	gaa 1	20
f	花	faa 1	18
s	沙	saa 1	15
h	蝦	haa 1	12
m	媽	maa 1	21
n	那	naa 5	19
ng	牙	ngaa 4	20
c	叉	caa 1	19 15
z	渣	zaa 1	19 15
l	啦	laa 1	14
kw	誇	kwaa 1	20
gw	瓜	gwaa 1	20
w	蛙	waa 1	7
j	廿	jaa 6	6

Jyutping Middle for zh-HK

Expand table

Jyutping	Character example	sapi example	VisemeID
aa	沙	saa 1	2
e	些	se 1	4
i	詩	si 1	6

Jyutping	Character example	sapi example	VisemeID
o	疏	so 1	3 In [ou], [o] corresponding viseme is 8
u	夫	fu 1	7
oe	鋸	goe 3	4
yu	書	syu 1	4
a	新	san 1	4
eo	律	leot 6	1

Jyutping Ending for zh-HK

Expand table

Jyutping	Character example	sapi example	VisemeID
i	西	sai 1	6 (In [eoi], [i] corresponding viseme is 4)
u	收	sau 1	7
p	夾	gep 6	21
t	不	bat 1	19
k	策	caak 3	20
m	心	sam 1	21
n	新	san 1	19
ng	敬	ging 3	20

Jyutping Tone for zh-HK

Expand table

Tone number	Description	Character example	sapi example
1	High level/High falling or Entering High Level	詩	si 1
2	Mid Rising	史	si 2
3	Mid Level or Entering Mid Level	試	si 3

Tone number	Description	Character example	sapi example
4	Low Falling	時	si 4
5	Low Rising	市	si 5
6	Low Level or Entering Low Level	是	si 6

zh-TW

The Speech service phone set for zh-TW is based on the native phone [Bopomofo](#) [↗](#) set.

Bopomofo Initials for zh-TW

[↗](#) Expand table

Bopomofo	Pinyin	sapi Example (Bopomofo, Pinyin)
ㄅ	b	玻 (ㄅㄛ, bo 1)
ㄆ	p	坡 (ㄆㄛ, po 1)
ㄇ	m	摸 (ㄇㄛ, mo 1)
ㄈ	f	佛 (ㄈㄛˊ, fo 2)
ㄉ	d	得 (ㄉㄛˊ, de 2)
ㄊ	t	特 (ㄊㄛˋ, te 4)
ㄋ	n	呢 (ㄋㄛˋ, ne 5)
ㄌ	l	樂 (ㄌㄛˋ, le 4)
ㄍ	g	哥 (ㄍㄛ, ge 1)
ㄎ	k	科 (ㄎㄛ, ke 1)
ㄏ	h	喝 (ㄏㄛ, he 1)
ㄐ	j	基 (ㄐㄧ, ji 1)
ㄑ	q	欺 (ㄑㄧ, qi 1)
ㄒ	x	希 (ㄒㄧ, xi 1)
ㄓ	zh	知 (ㄓ, zhi 1)

Bopomofo	Pinyin	sapi Example (Bopomofo, Pinyin)
ㄷ	ch	吃 (ㄷ, chi 1)
ㄕ	sh	詩 (ㄕ, shi 1)
ㄣ	r	日 (ㄣ, ri 4)
ㄗ	z	資 (ㄗ, zi 1)
ㄘ	c	此 (ㄘ, ci 3)
ㄙ	s	思 (ㄙ, si 1)
ㄣ	y	衣 (ㄣ, yi 1)
ㄨ	w	屋 (ㄨ, wu 1)

Bopomofo Finals for zh-TW

 Expand table

Bopomofo	Pinyin	sapi Example (Bopomofo, Pinyin)
ㄚ	a	法 (ㄘ ㄚ, fa 3)
ㄛ	o	潑 (ㄗ ㄛ, po 1)
ㄜ	e	歌 (ㄍ ㄜ, ge 1)
ㄝ	i	理 (ㄌ ㄝ, li 3)
ㄨ	u	步 (ㄅ ㄨ, bu 4)
ㄣ	v	女 (ㄋ ㄣ, nv 3)
ㄞ	ai	百 (ㄅ ㄞ, bai 3)
ㄟ	ei	北 (ㄅ ㄟ, bei 3)
ㄠ	ui	對 (ㄉ ㄠ, dui 4)
ㄡ	ao	號 (ㄏ ㄡ, hao 4)
ㄣ	ou	走 (ㄗ ㄣ, zou 3)
ㄤ	iu	牛 (ㄋ ㄤ, niu 2)
ㄤ	ie	謝 (ㄒ ㄤ, xie 4)

Bopomofo	Pinyin	sapi Example (Bopomofo, Pinyin)
ㄌㄝˊ	ue	略 (ㄌㄌㄝˊ, lue 4)
ㄏㄢˋ	an	喊 (ㄏㄢˋ, han 3)
ㄎㄣˋ	en	肯 (ㄎㄎㄣˋ, ken 3)
ㄅㄧㄣˊ	in	賓 (ㄅㄅㄧㄣˊ, bin 1)
ㄗㄨㄣˊ	un	尊 (ㄗㄗㄨㄣˊ, zun 1)
ㄌㄤˋ	ang	朗 (ㄌㄌㄤˋ, lang 3)
ㄏㄥˊ	eng	恆 (ㄏㄥˊ, heng 2)
ㄩㄥˊ	ing	贏 (ㄩㄥˊ, ying 2)
ㄏㄨㄥˊ	ong	紅 (ㄏㄥˊ, hong 2)
ㄐㄧㄚˊ	ia	家 (ㄐㄐㄧㄚˊ, jia 1)
ㄇㄧㄢˋ	ian	麵 (ㄇㄐㄧㄢˋ, mian 4)
ㄒㄧㄤˋ	iang	象 (ㄒㄐㄧㄤˋ, xiang 4)
ㄅㄧㄠˋ	iao	表 (ㄅㄐㄧㄠˋ, biao 3)
ㄒㄩㄥˊ	iong	兄 (ㄒㄐㄩㄥˊ, xiong 1)
ㄏㄨㄚˊ	ua	花 (ㄏㄥˊ, hua 1)
ㄕㄨㄞˋ	uai	帥 (ㄕㄕㄨㄞˋ, shuai 4)
ㄉㄨㄢˋ	uan	短 (ㄉㄐㄨㄢˋ, duan 3)
ㄍㄨㄤˋ	uang	廣 (ㄍㄕㄨㄤˋ, guang 3)
ㄉㄨㄛˊ	uo	多 (ㄉㄐㄨㄛˊ, duo 1)

Bopomofo Whole syllable zh-TW

[Expand table](#)

Bopomofo	Pinyin	sapi Example (Bopomofo, Pinyin)
ㄓ	zhi	知 (ㄓ, zhi 1)
ㄔ	chi	吃 (ㄔ, chi 1)

Bopomofo	Pinyin	sapi Example (Bopomofo, Pinyin)
尸	shi	诗 (尸, shi 1)
日	ri	日 (日, ri 1)
ㄗ	zi	资 (ㄗ, zi 1)
ㄘ	ci	词 (ㄘ, ci 2)
ㄌ	si	斯 (ㄌ, si 1)
ㄌ	yi	衣 (ㄌ, yi 1)
ㄨㄛ	wo	我 (ㄨㄛ, wo 3)
ㄨ	wu	屋 (ㄨ, wu 1)
ㄩ	yu	鱼 (ㄩ, yu 2)
ㄌ ㄝ	ye	叶 (ㄌ ㄝ, ye 4)
ㄩ ㄝ	yue	月 (ㄩ ㄝ, yue 4)
ㄦ	er	耳 (ㄦ, er 3)
ㄩ ㄢ	yuan	圆 (ㄩ ㄢ, yuan 2)
ㄌ ㄣ	yin	音 (ㄌ ㄣ, yin 1)
ㄩ ㄣ	yun	云 (ㄩ ㄣ, yun 1)
ㄌ ㄥ	ying	英 (ㄌ ㄥ, ying 1)
ㄩ ㄥ	yong	拥 (ㄩ ㄥ, yong 1)
ㄚ	a	啊 (ㄚ, a 1)
ㄛ	o	噢 (ㄛ, o 1)
ㄜ	e	鹅 (ㄜ, e 2)
ㄞ	ai	爱 (ㄞ, ai 4)
ㄟ	ei	欸 (ㄟ, ei 1)
ㄠ	ao	奥 (ㄠ, ao 4)
ㄡ	ou	偶 (ㄡ, ou 3)
ㄢ	an	安 (ㄢ, an 1)
ㄣ	en	恩 (ㄣ, en 1)

Bopomofo	Pinyin	sapi Example (Bopomofo, Pinyin)
ㄤ	ang	昂 (ㄤˊ, ang 2)
ㄥ	eng	鞞 (ㄥ, eng 1)
ㄝ	ê	ㄝ (ㄝ, ê 1)

Bopomofo tone zh-TW

[🔗](#) Expand table

Bopomofo	Tone number	sapi Example (Bopomofo, Pinyin)
ˊ	1	八 (ㄅㄚˊ or ㄅㄚˇ, ba 1)
ˊˊ	2	拔 (ㄅㄚˊˊ, ba 2)
ˇ	3	把 (ㄅㄚˇ, ba 3)
ˋ	4	坝 (ㄅㄚˋ, ba 4)
ˊˋ	5	吧 (ㄅㄚˊˋ, ba 5)

Map X-SAMPA to IPA

When using X-SAMPA phone symbols `'` and `"`, it's important to avoid conflicts with the wrapper symbol. If the X-SAMPA string contains phone `'`, we recommend using phone `_j` instead. If you don't want to replace the phone `'`, you need to use double quotes `"` as a wrapper. Similarly, if the X-SAMPA string contains phone `"`, then double quotes shouldn't be used as a wrapper, and you must use the single quote `'` as a wrapper. Otherwise, it will cause an error.

The table below shows a mapping relationship between X-SAMPA (Extended Speech Assessment Methods Phonetic Alphabet) and IPA alphabets. The X-SAMPA symbols are shown at left, with the corresponding IPA symbols to the right.

txt	
x-sampa (L) ipa (R) a a b b b_< b̥	

c
d
d`
d_<
e
f
g
g_<
h
h\
i
j
j\
k
l
l`
l\
m
n
n`
o
p
p\
q
r
r`
r\
r\`
s
s`
s\
t
t`
u
v
P
v\
w
x
x\
y
z
z`
z\
A
B
B\
C
D
E
F
G
G\
G_<
H

c
d
d
d
e
f
g
g
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u
w
x
h
y
z
z
z
a
b
B
c
d
e
e
Y
G
G
y

H\	H
I	I
I\	ɪ
J	ɲ
J\	ʝ
J_<	ʝ
K	ɿ
K\	ʔ
L	ʌ
L\	L
M	ʍ
M\	ɰ
N	ŋ
N\	N
O	ɔ
O\	θ
Q	ɒ
R	ʁ
R\	R
S	ʃ
T	θ
U	ʊ
U\	ʉ
V	ʌ
W	M
X	χ
X\	ħ
Y	ʏ
Z	ʒ
.	.
„	‚
%	%
_j	˙
˙	˙
:	:
:\	˙
@	ə
@\	ə
@`	ə
{	æ
}	ʈ
1	ɿ
2	∅
3	ɜ
3\	ɐ
4	ɾ
5	ɿ
6	e
7	ʏ
8	θ
9	œ
&	œ
?	ʔ
?\	ʔ
<\	ʔ

_w
_X
_X

w
~
x

Lower speech synthesis latency using Speech SDK

08/07/2025

In this article, we introduce the best practices to lower the text to speech synthesis latency and bring the best performance to your end users.

Normally, we measure the latency by `first byte latency` and `finish latency`, as follows:

 Expand table

Latency	Description	<code>SpeechSynthesisResult</code> property key
<code>first byte client latency</code>	Indicates the time delay between the synthesis starts and the first audio chunk is received on the client including network latency.	<code>SpeechServiceResponse_SynthesisFirstByteLatencyMs</code>
<code>finish client latency</code>	Indicates the time delay between the synthesis starts and the whole synthesized audio is received on the client including network latency.	<code>SpeechServiceResponse_SynthesisFinishLatencyMs</code>
<code>network latency</code>	The network latency between the client and Azure TTS service.	<code>SpeechServiceResponse_SynthesisNetworkLatencyMs</code>
<code>first byte service latency</code>	Indicates the time delay between Azure TTS service received synthesis request and the first audio chunk is returned.	<code>SpeechServiceResponse_SynthesisServiceLatencyMs</code>

The Speech SDK puts the latency durations in the Properties collection of `SpeechSynthesisResult`. The following sample code shows these values.

C#

```
var result = await synthesizer.SpeakTextAsync(text);
Console.WriteLine($"first byte client latency:
\t{result.Properties.GetProperty(PropertyId.SpeechServiceResponse_SynthesisFirstByteLatencyMs)} ms");
Console.WriteLine($"finish client latency:
\t{result.Properties.GetProperty(PropertyId.SpeechServiceResponse_SynthesisFinishLatencyMs)} ms");
Console.WriteLine($"network latency:
\t{result.Properties.GetProperty(PropertyId.SpeechServiceResponse_SynthesisNetworkLatencyMs)} ms");
Console.WriteLine($"first byte service latency:
```

```
\t{result.Properties.GetProperty(PropertyId.SpeechServiceResponse_SynthesisServiceLatencyMs)} ms");  
// you can also get the result id, and send to us when you need help for diagnosis  
var resultId = result.ResultId;
```

The first byte latency is lower than finish latency in most cases. The first byte latency is independent from text length, while finish latency increases with text length.

Ideally, we want to minimize the user-experienced latency (the latency before user hears the sound) to one network route trip time plus the first audio chunk latency of the speech synthesis service.

Streaming

Streaming is critical to lowering latency. Client code can start playback when the first audio chunk is received. In a service scenario, you can forward the audio chunks immediately to your clients instead of waiting for the whole audio.

You can use the [PullAudioOutputStream](#), [PushAudioOutputStream](#), [Synthesizing event](#), and [AudioDataStream](#) of the Speech SDK to enable streaming.

Taking `AudioDataStream` as an example:

C#

```
using (var synthesizer = new SpeechSynthesizer(config, null as AudioConfig))  
{  
    using (var result = await synthesizer.StartSpeakingTextAsync(text))  
    {  
        using (var audioDataStream = AudioDataStream.FromResult(result))  
        {  
            byte[] buffer = new byte[16000];  
            uint filledSize = 0;  
            while ((filledSize = audioDataStream.ReadData(buffer)) > 0)  
            {  
                Console.WriteLine($"{filledSize} bytes received.");  
            }  
        }  
    }  
}
```

Pre-connect and reuse SpeechSynthesizer

The Speech SDK uses a websocket to communicate with the service. Ideally, the network latency should be one route trip time (RTT). If the connection is newly established, the network

latency includes extra time to establish the connection. The establishment of a websocket connection needs the TCP handshake, SSL handshake, HTTP connection, and protocol upgrade, which introduces time delay. To avoid the connection latency, we recommend pre-connecting and reusing the `SpeechSynthesizer`.

Pre-connect

To pre-connect, establish a connection to the Speech service when you know the connection is needed soon. For example, if you're building a speech bot in client, you can pre-connect to the speech synthesis service when the user starts to talk, and call `SpeakTextAsync` when the bot reply text is ready.

C#

```
using (var synthesizer = new SpeechSynthesizer(uspConfig, null as AudioConfig))
{
    using (var connection = Connection.FromSpeechSynthesizer(synthesizer))
    {
        connection.Open(true);
    }
    await synthesizer.SpeakTextAsync(text);
}
```

ⓘ Note

If the text is available, just call `SpeakTextAsync` to synthesize the audio. The SDK will handle the connection.

Reuse SpeechSynthesizer

Another way to reduce the connection latency is to reuse the `SpeechSynthesizer` so you don't need to create a new `SpeechSynthesizer` for each synthesis. We recommend using object pool in service scenario. See our sample code for [C#](#) and [Java](#).

Transmit compressed audio over the network

When the network is unstable or with limited bandwidth, the payload size also affects latency. Meanwhile, a compressed audio format helps to save the users' network bandwidth, which is especially valuable for mobile users.

We support many compressed formats including `opus`, `webm`, `mp3`, `silk`, and so on, see the full list in [SpeechSynthesisOutputFormat](#). For example, the bitrate of `Riff24Khz16BitMonoPcm` format is 384 kbps, while `Audio24Khz48KBitRateMonoMp3` only costs 48 kbps. The Speech SDK automatically uses a compressed format for transmission when a `pcm` output format is set. For Linux and Windows, `GStreamer` is required to enable this feature. Refer [this instruction](#) to install and configure `GStreamer` for Speech SDK. For Android, iOS, and macOS, no extra configuration is needed starting version 1.20.

Input text streaming

Text streaming allows real-time text processing for rapid audio generation. It's perfect for dynamic text vocalization, such as reading outputs from AI models like GPT in real-time. This feature minimizes latency and improves the fluidity and responsiveness of audio outputs, making it ideal for interactive applications, live events, and responsive AI-driven dialogues.

How to use text streaming

Text streaming is supported in C#, C++ and Python with Speech SDK.

To use the text streaming feature, connect to the websocket V2 endpoint:

```
wss://{region}.tts.speech.microsoft.com/cognitiveservices/websocket/v2
```

See the sample code for setting the endpoint:

C#

```
// IMPORTANT: MUST use the websocket v2 endpoint
var ttsEndpoint =
    $"wss://{Environment.GetEnvironmentVariable("AZURE_TTS_REGION")}.tts.speech.microsoft.com/cognitiveservices/websocket/v2";
var speechConfig = SpeechConfig.FromEndpoint(
    new Uri(ttsEndpoint),
    Environment.GetEnvironmentVariable("AZURE_TTS_API_KEY"));
```

Key steps

1. **Create a text stream request:** Use `SpeechSynthesisRequestInputType.TextStream` to initiate a text stream.
2. **Set global properties:** Adjust settings such as output format and voice name directly, as the feature handles partial text inputs and doesn't support SSML. Refer to the following sample code for instructions on how to set them. OpenAI text to speech voices aren't

supported by the text streaming feature. See this [language table](#) for full language support.

C#

```
// Set output format
speechConfig.SetSpeechSynthesisOutputFormat(SpeechSynthesisOutputFormat.Raw24
Khz16BitMonoPcm);

// Set a voice name
SpeechConfig.SetProperty(PropertyId.SpeechServiceConnection_SynthVoice, "en-
US-AvaMultilingualNeural");
```

3. **Stream your text:** For each text chunk generated from a GPT model, use `request.InputStream.Write(text);` to send the text to the stream.
4. **Close the stream:** Once the GPT model completes its output, close the stream using `request.InputStream.Close();`.

For detailed implementation, see the [sample code on GitHub](#) [↗](#)

Others tips

Cache CRL files

The Speech SDK uses CRL files to check the certification. Caching the CRL files until expired helps you avoid downloading CRL files every time. See [How to configure OpenSSL for Linux](#) for details.

Use latest Speech SDK

We keep improving the Speech SDK's performance, so try to use the latest Speech SDK in your application.

Load test guideline

You can use load test to test the speech synthesis service capacity and latency. Here are some guidelines:

- The speech synthesis service has the ability to autoscale, but takes time to scale out. If the concurrency is increased in a short time, the client might get long latency or `429` error

code (too many requests). So, we recommend you increase your concurrency step by step in load test. [See this article](#) for more details, especially [this example of workload patterns](#).

- You can use our sample using object pool ([C#](#) and [Java](#)) for load test and getting the latency numbers. You can modify the test turns and concurrency in the sample to meet your target concurrency.
- The service has quota limitation based on the real traffic, therefore, if you want to perform load test with the concurrency higher than your real traffic, connect before your test.

Next steps

- [See the samples](#) on GitHub

Get facial position with viseme

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ⓘ Note

To explore the locales supported for viseme ID and blend shapes, refer to [the list of all supported locales](#). Scalable Vector Graphics (SVG) is only supported for the `en-US` locale.

A *viseme* is the visual description of a phoneme in spoken language. It defines the position of the face and mouth while a person is speaking. Each viseme depicts the key facial poses for a specific set of phonemes.

You can use visemes to control the movement of 2D and 3D avatar models, so that the facial positions are best aligned with synthetic speech. For example, you can:

- Create an animated virtual voice assistant for intelligent kiosks, building multi-mode integrated services for your customers.
- Build immersive news broadcasts and improve audience experiences with natural face and mouth movements.
- Generate more interactive gaming avatars and cartoon characters that can speak with dynamic content.
- Make more effective language teaching videos that help language learners understand the mouth behavior of each word and phoneme.
- People with hearing impairment can also pick up sounds visually and "lip-read" speech content that shows visemes on an animated face.

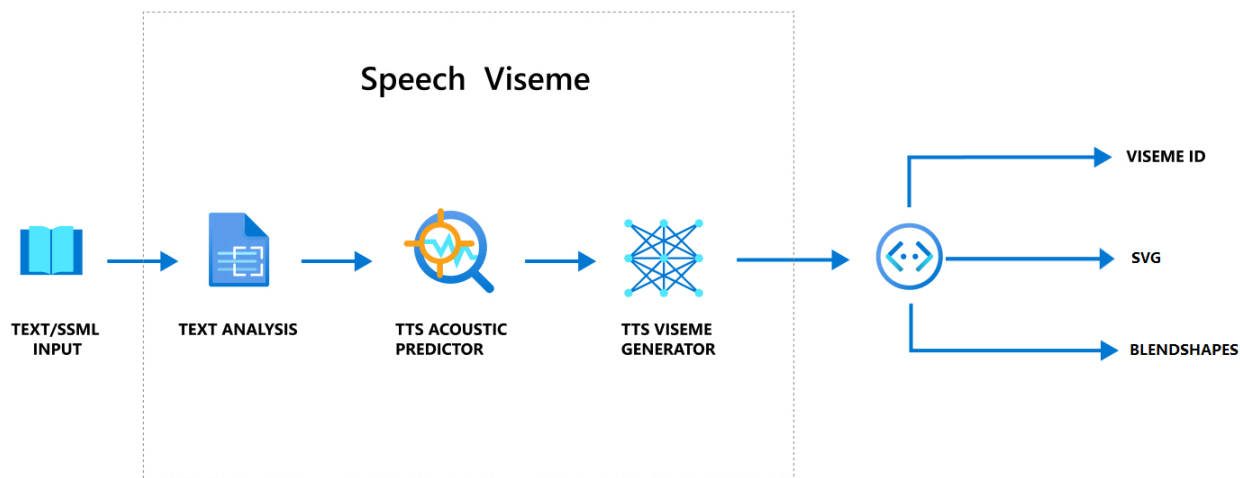
For more information about visemes, view this [introductory video](#) .

<https://www.youtube-nocookie.com/embed/ui9XT47uwxs> 

Overall workflow of producing viseme with speech

Neural Text to speech (Neural TTS) turns input text or SSML (Speech Synthesis Markup Language) into lifelike synthesized speech. Speech audio output can be accompanied by viseme ID, Scalable Vector Graphics (SVG), or blend shapes. Using a 2D or 3D rendering engine, you can use these viseme events to animate your avatar.

The overall workflow of viseme is depicted in the following flowchart:



Viseme ID

Viseme ID refers to an integer number that specifies a viseme. We offer 22 different visemes, each depicting the mouth position for a specific set of phonemes. There's no one-to-one correspondence between visemes and phonemes. Often, several phonemes correspond to a single viseme, because they looked the same on the speaker's face when they're produced, such as **s** and **z**. For more specific information, see the table for [mapping phonemes to viseme IDs](#).

Speech audio output can be accompanied by viseme IDs and **Audio offset**. The **Audio offset** indicates the offset timestamp that represents the start time of each viseme, in ticks (100 nanoseconds).

Map phonemes to visemes

Visemes vary by language and locale. Each locale has a set of visemes that correspond to its specific phonemes. The [SSML phonetic alphabets](#) documentation maps viseme IDs to the corresponding International Phonetic Alphabet (IPA) phonemes. The table in this section shows a mapping relationship between viseme IDs and mouth positions, listing typical IPA phonemes for each viseme ID.

[Expand table](#)

Viseme ID	IPA	Mouth position
0	Silence	
1	æ, ə, ʌ	
2	ɑ	
3	ɔ	

Viseme ID	IPA	Mouth position
4	ɛ, ʊ	
5	ə	
6	j, i, ɪ	
7	w, u	